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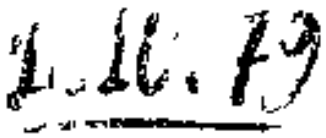
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Chapter 16

GELFAND-P000848

[source number: 1]

1.10.79 Gelfand Sem.



Original source details: GELFAND-P000848

Uncertain reading. The month is provisionally read as “10.” The second digit is incomplete in the scan.

[Crossed-out heading: ~~KALLOSH?~~]



Original source details: GELFAND-P000848

Uncertain reading. «КАЛЛОШ» (“KALLOSH”) is a possible reading of the crossed-out heading.

I. M. G. What is a spinor?

Pauli’s book on quant. mech. — the best expos. of Dirac’s theory (Dirac eq.)



Original source details: GELFAND-P000848

Uncertain reading. The overwritten abbreviation is provisionally read «квант.» (“quant.”).

His objections to Dirac’s theor.:

a) exist. of positrons, predicted by Dirac’s theor.

b) if they exist, then on colliding with a positr. one gets 2 photons



Original source details: GELFAND-P000848

Uncertain reading. The abbreviation after “with” is provisionally read «позитр.» (“positr.”).

c) according to this theory 2 photons can produce an electr./positron pair



Original source details: GELFAND-P000848

Uncertain reading. The abbreviation before the slash is tentatively read «эл-н» (“electr.”).

Dirac refus., so that they would not make him “Sir” (?)



Original source details: GELFAND-P000848

Uncertain reading. The abbreviation after “Dirac” is provisionally read «отказ.» (“refus.”).

[The lines are grouped by a bracket on the left.]

Once someone told him that he did not understand the form. in the lower left corner of the board.

Dirac does not answer.

The chairman asks him to answer

[Crossed out: ...] Dirac: “there was no question, there was a statement.”



Original source details: GELFAND-P000848

Undeciphered passage. The crossed-out fragment before “Dirac” is illegible.



Original source details: GELFAND-P000848

Diagram and typed description. 
Bracket to the left of the preceding four lines.

Hamilton: wanted to extract the root of $x^2 + y^2 + z^2$, and obtain a “lin. form”

$$-(x^2 + y^2 + z^2) = (ax + by + cz)^2$$

$$a^2 = b^2 = c^2 = -1$$



Original source details: GELFAND-P000848

Uncertain reading. The overwritten sign after c^2 is provisionally read $= -1$.

$$(ab + ba)\,[?]\,xy + (ac + ca)xz + (bc + cb)yz$$
$$+ a^2x^2 + b^2y^2 + c^2z^2 = -(x^2 + y^2 + z^2).$$



Original source details: GELFAND-P000848

Uncertain reading. There is an unresolved dark stroke between the first parenthesis and xy.

hence the relations

$$ab + ba = 0 \quad \text{etc.}$$

Add the unit — get quaternions.

Consider the transformation

$$z' = \frac{az + b}{cz + d}$$

or

$$\begin{cases} z_1' = az_1 + bz_2 \\ z_2' = cz_1 + dz_2 \end{cases} \qquad z' = \frac{z_1'}{z_2'}, \quad z = \frac{z_1}{z_2}.$$

Spinors — a pair of numbers [below "numbers": (compl.)], transf. according to this law

Compl. conj. — conj. spinor.



Diagram and typed description.

—— • ——

Two short strokes with a dot between them.

Original source details: GELFAND-P000848

Gravitation should be read in Landau and Lifshitz (vol. 2)

chs. 1, 2, 10, 11.

Renata E. Kallosh.

Program.

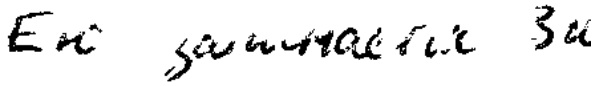
1. Two-component spinors.
2. Self-duality.
3. Diff. operator $\Delta(A, B)$, descr. the eq. for particles of higher spins.
4. Heat eq., zero and nonzero modes.
5. Index theorems.
6. Supertheor. in a grav. field
7. Superself-duality.

(8. Index supertheor.! Singer is working on it, but in fact there is no such theorem)



Uncertain reading. The numeral after the opening parenthesis is tentatively read as 8.

Original source details: GELFAND-P000848



Uncertain reading. The opening phrase is provisionally read «Ею занимается» ("is working on it").

Original source details: GELFAND-P000848

GELFAND-P000849

Refs.

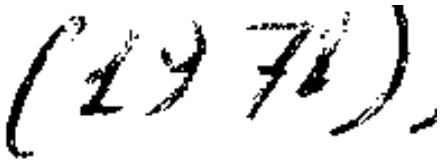
1. [Crossed out: ...] S. W. Hawking, C. N. Pope



Undeciphered passage. A short fragment before “S. W. Hawking” is crossed out and illegible.

Original source details: GELFAND-P000849

Nucl. Phys., B 146 (1978), 381



Uncertain reading. The final digit of the year is provisionally read as 8. The strokes could also be read as 1.

Original source details: GELFAND-P000849

2. R. E. Kallosh, JETP Letters,

29 (1979), 192; 493

3. S. M. Christensen, M. J. Duff.

Nucl. Phys. (1979), 301

4. — „ —, — „ — Deser, Gr[?]

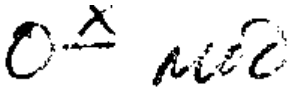


Uncertain reading. The name after “Deser” is provisionally read “Gr” with an unclear ending. The first letter could be read as “C.”

Original source details: GELFAND-P000849

Phys Lett. 84 B (1979), 411

[1] — main contrib. in supergrav. (in the 1-loop approx.) — from $0^{\underline{x}}$ modes



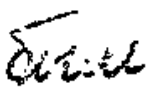
Uncertain reading. There is a small underlined superscript «x» (the Russian ordinal ending) at the zero.

Original source details: GELFAND-P000849

(in Y.-M. — from all modes!)

[2], [3] indep. — in supergrav. (for self-dual) for higher orders

Before this it was so in the 1st and 2nd ord. supergrav. did not cont. diverg.



Uncertain reading. The word after “Before this” is provisionally read «было» (“it was”). The last letter is unclear.

Original source details: GELFAND-P000849

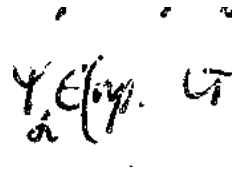
[3] based on the ordinary index theor.

Two-comp. van der Waerden formalism.

$$\Psi_\alpha \longrightarrow \Psi_\beta \Omega_\alpha^\beta \tag{1}$$

[At right:] Transformation law

$\Psi_\alpha \in$ (repr. stand. 2-dim. repr. $SL(2,C)) = V$;

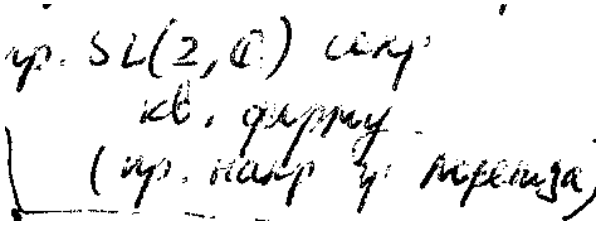


Original source details: GELFAND-P000849

Uncertain reading. The first abbreviation is provisionally read «пр.» (“repr.”). Its initial letter could be «и».

$\Psi_{\dot\alpha} \in \bar V$, $V \otimes \bar V$ has a real str. and the gr. $SL(2,C)$ pres. the quad. form.

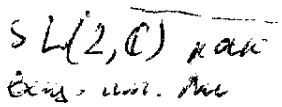
(gr. e.g. in Penrose)



Original source details: GELFAND-P000849

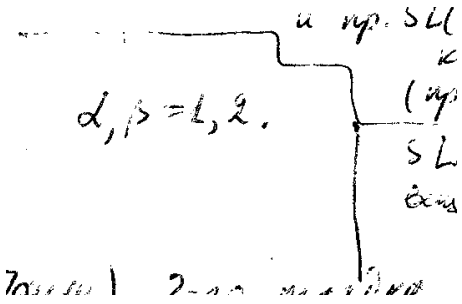
Uncertain reading. The abbreviation in parentheses is provisionally read «напр.» (“e.g.”).

$SL(2,C)$ as a real Lie man.



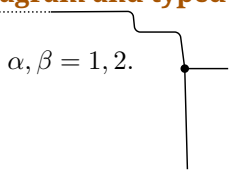
Original source details: GELFAND-P000849

Uncertain reading. The abbreviation after “real” is provisionally read «мн.» (“man.”). «гр.» (“gr.”) is possible.



Original source details: GELFAND-P000849

Diagram and typed description.



A stepped line between the index formula and the note at right. A short line extends right from the dot. The correspondence between strokes and lines of text is unclear.

[Main text:] If the Lorentz gr. or $SO(4)$

$$\Omega_\alpha^\beta = (\omega^{ab}\Sigma_{ab})_\alpha^\beta \tag{2}$$

$\alpha, \beta = 1, 2.$

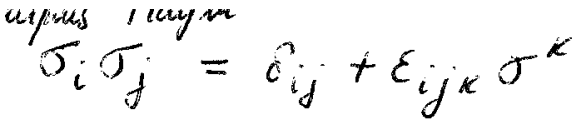
(transf. law of a two-comp. spinor)

Σ_{ab} — fixed matrices (like the Pauli matr.) of order 2

ω^{ab} — parameters

For the Pauli matrices

$$\sigma_i \sigma_j = \delta_{ij} + \epsilon_{ijk}\sigma^k \tag{3}$$



Uncertain reading. The letter ϵ is visible after the plus sign in (3). No separate factor precedes it.

Original source details: GELFAND-P000849

$$\sigma^a = (I, i\vec{\sigma}), \qquad a = 1, 2, 3, 4.$$

$$\Sigma_{ab} = \sigma_a \bar{\sigma}_b - \bar{\sigma}_b \sigma_a \tag{4}$$

Now about quantities in spinor notation.

Penrose formulated the theory of gravitation in this way.

Let there be a vector x_a; then

$$X_{\alpha\dot{\alpha}} = \sigma^a_{\alpha\dot{\alpha}} x_a \quad \text{— spinor} \tag{5}$$

$$\Psi_{\alpha_1 \ldots \alpha_{2A}, \dot{\alpha}_1 \ldots \dot{\alpha}_{2B}} = \Psi(A, B) \tag{6}$$



Uncertain reading. The first dotted index is provisionally set as $\dot{\alpha}_1$. The numeral can also be read as 2.

Original source details: GELFAND-P000849

[source number: 2]

and if one fully symmetrizes, one gets an irred. representation.
Incidentally, about the Pauli matrices: they are as follows.

$$\begin{cases} \sigma^1_{\alpha\dot\alpha} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \\ \sigma^2_{\alpha\dot\alpha} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \\ \sigma^3_{\alpha\dot\alpha} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}; \\ \sigma^4_{\alpha\dot\alpha} = \dfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{cases} \tag{7}$$



Uncertain reading. The first lower index of σ^1 is overwritten. The upper layer reads α. The earlier sign is illegible.

Original source details: GELFAND-P000850

There is also $\epsilon^{\alpha\beta}$

$$\epsilon^{12} = 1, \qquad \epsilon_{\alpha\beta} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \tag{8}$$

Lowering indices:

$$\Psi_\alpha = \epsilon_{\alpha\beta}\Psi^\beta \tag{9}$$

I. M. G.
In coord. $\dot\alpha$ the transf. is complex conjugate.
Form. (5) — constr. of a spinor from a vector



Diagram and typed description. (!)
Circled exclamation mark to the left of the line about formula (5).

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Diagram and typed description. — . ——

Original source details: GELFAND-P000850

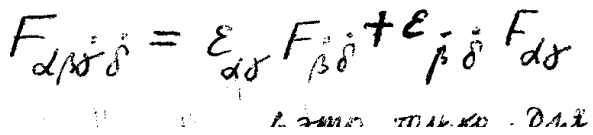
K. Consid. [crossed out: ...] now antisymmetr. tensors



Undeciphered passage. The word crossed out before “now” is illegible.

Original source details: GELFAND-P000850

$$F_{[ab]} \longmapsto F_{\alpha\beta\dot\gamma\dot\delta} = \epsilon_{\alpha\gamma} F_{\dot\beta\dot\delta} + \epsilon_{\dot\beta\dot\delta} F_{\alpha\gamma}, \tag{10}$$

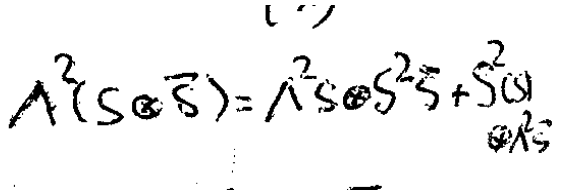


Uncertain reading. On the left of the equals sign the dots are visible above the third and fourth indices. On the right the dotted indices are provisionally read as β, δ.

Original source details: GELFAND-P000850

[Below $F_{[ab]}$:] antisymm. tensor
(this is only for antisymm.)
— easy to check
[At right:] $F_{\alpha\gamma}, F_{\dot\beta\dot\delta}$ — symm. tensors
[In the right margin:]

$$\Lambda^2(S \otimes \bar S) = \Lambda^2 S \otimes S^2 \bar S + S^2(S) \otimes \Lambda^2 \bar S \tag{*}$$



Uncertain reading. The second summand continues on a new line. The S after S^2 appears to be in parentheses.

Original source details: GELFAND-P000850

eigenval. of oper. $*$ — ± 1
Self-dual tensor: def. oper. $*$:

$${}^*F_{ab} = \frac{1}{2}\epsilon_{abcd} F^{cd} \tag{11}$$

self-duality:

$$F_{ab} = \pm {}^*F_{ab}; \quad \text{[crossed out: ...]} \quad \{F_{\alpha\beta} = 0 \quad \text{or} \quad F_{\dot\alpha\dot\beta} = 0\} \tag{12}$$



Undeciphered passage. The crossed-out passage before the brace is illegible.

Original source details: GELFAND-P000850

On gravitation

Curvature tensor: R_{abcd}. A metric is assumed to exist
and definition of the operation $*$:

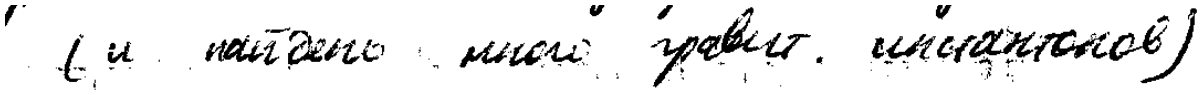
$$ {}^*R_{abcd} = \frac{1}{2}\epsilon_{ab}{}^{c'd'} R_{c'd'cd} \tag{13} $$



Undeciphered passage. Faint marks are visible to the right of (13). They do not yield a reading.

Original source details: GELFAND-P000850

Spinor equivalent. Einste. eqs. follow from the self-duality cond.
(and many grav. instantons have been found)



Original source details: GELFAND-P000850

Uncertain reading. The Russian word after “and” is provisionally read «найдено» (“have been found”).

$$ R_{\alpha\beta\gamma\delta} \quad \text{and} \quad R_{\dot\alpha\dot\beta\dot\gamma\dot\delta} $$

$$ R_+ = R + {}^*R = 0 \quad \text{— see eq. } (*). $$

[At right:] see Hawking’s lect., 1978 with a reference to Penrose



Diagram and typed description. Vertical mark with transverse strokes to the left of the reference to the lectures.

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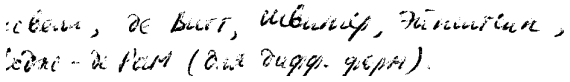
$$ \Phi(A,B) = \Psi_{\alpha_1,\ldots,\alpha_{2A},\dot\alpha_1,\ldots,\dot\alpha_{2B}} \tag{14} $$



Uncertain reading. The letter in (14) is set as Φ. It is unclear whether the top stroke is a separate bar.

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The eqs. cont. a bunch of different eqs. — Maxwell, de Wit, Schwinger, Einstein, Hodge–de Rham (for diff. forms).



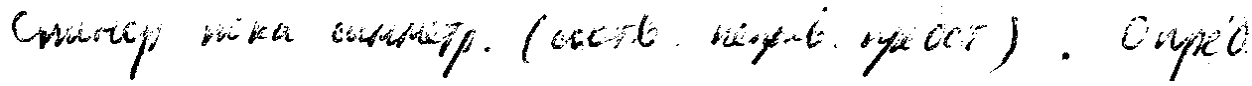
Uncertain reading. The surname after “de Wit” is provisionally read «Швингер» (“Schwinger”).

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(see [3])

GELFAND-P000851

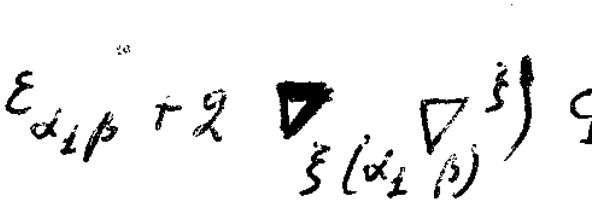
The spinor is symmetr. for now (corresp. irred. repr.). Definition of Δ.



Original source details: GELFAND-P000851

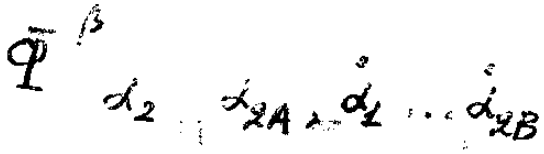
Uncertain reading. The abbreviation before “irred.” is provisionally read «соотв.» (“corresp.”).

$$(\Delta\phi)_{\alpha_1\dots\alpha_{2A},\dot\alpha_1\dots\dot\alpha_{2B}} = \big(-\Box\epsilon_{\alpha_1\beta} + 2\nabla_{\dot\xi(\alpha_1}\nabla^{\dot\xi}_{\beta)}\big) \bar\phi^{\beta}{}_{\alpha_2\dots\alpha_{2A},\dot\alpha_1\dots\dot\alpha_{2B}}. \tag{15}$$



Original source details: GELFAND-P000851

Uncertain reading. The small dots in the contraction are provisionally assigned to the two ξ indices. Parentheses enclose the lower α_1, β.



Original source details: GELFAND-P000851

Uncertain reading. The right-hand side of (15) is provisionally set with $\bar\phi^{\beta}$. A short bar is visible above the letter.

Properties.

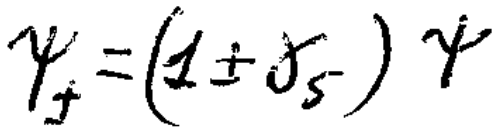
$$1) \qquad (\phi, \Delta\phi) = (\Delta\phi, \phi) \tag{16}$$

$$2) \qquad \Delta\phi(0,0) = -\Box\phi, \qquad \text{here} \quad \Box = \nabla_{\alpha\dot\alpha}\nabla^{\alpha\dot\alpha}, \tag{17}$$

$$\nabla_{\alpha\dot\alpha} = \sigma^a_{\alpha\dot\alpha}\mathcal{D}_a; \qquad \mathcal{D}_a = \partial_a + B_a$$

3) Dirac $\phi(1/2, 0)$ (in curved sp.)

$$\Psi_\pm = (1 \pm \gamma_5)\Psi, \tag{18}$$



Original source details: GELFAND-P000851

Uncertain reading. A short horizontal stroke follows 5. Its function is unclear.

Ψ — 4-compon. Dirac spinor

$$\Delta\Psi_{\pm} = \left(-\Box + \frac{1}{4}R\right)\Psi_{\pm}$$

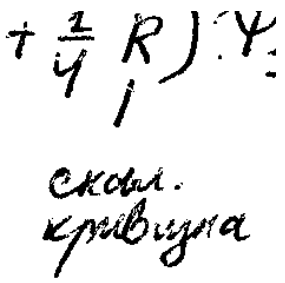


Diagram and typed description. R / scal. curvature
A stroke beneath R points to the label.

Original source details: GELFAND-P000851

4) Maxwell in harm. gauge



Uncertain reading. The abbreviation before “gauge” is provisionally read «гарм.» (“harm.”).

Original source details: GELFAND-P000851

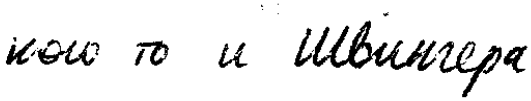
$$\Delta A_a = (-\Box g_{ab} + R_{ab})A^b, \qquad \mathcal{D}^a A_a = 0 \tag{19}$$



Uncertain reading. The operator before A_a is heavily overwritten. It is provisionally set as $\mathcal{D}$. ∇ is possible.

Original source details: GELFAND-P000851

5) $\underline{s = \frac{3}{2}}$ somebody and Schwinger (20)



Uncertain reading. The opening words are provisionally read «кого то» (“somebody”). The name before “and Schwinger” is left unspecified.

Original source details: GELFAND-P000851

6) $\underline{s = 2}$ Einstein (21)

7) $$w = \frac{1}{p!}\omega_{\mu_1\ldots\mu_p}\, dx^{\mu_1} \wedge \ldots \wedge dx^{\mu_p} \tag{22}$$



Uncertain reading. The sign left of the equals sign resembles Latin w. The indexed symbol at right is ω.

Original source details: GELFAND-P000851

$$\phi(0,0), \quad \phi(1/2,1/2), \quad \phi(1,0), \quad \phi(0,1)$$

$$\Delta = \delta d + d\delta$$

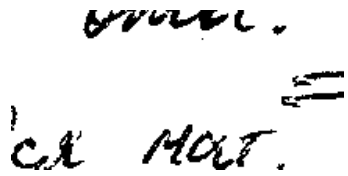
[crossed out: ...] Stand. theory of diff. forms descr. bosons.



Undeciphered passage. A sign before “Stand.” is crossed out. Its original form is illegible.

Original source details: GELFAND-P000851

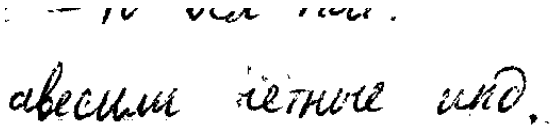
If there were odd ones — then all of math.



Uncertain reading. The line ends with «мат.» (“math.”) and breaks off.

Original source details: GELFAND-P000851

[3] — they attached even ind. to the form.



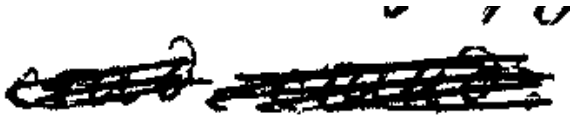
Uncertain reading. The last adjective is provisionally read «чётные» (“even”). Its opening strokes are faint.

Original source details: GELFAND-P000851

Needed: make anticommuting dx.

[At right:] Berezin did part of it.

There are as yet no [crossed out: ... ...]



Undeciphered passage. The two crossed-out words after “no” are illegible.

Original source details: GELFAND-P000851

diff. forms cont. among the dx comm. and anticomm. ones

introduce the operator Δ and construct an index theor.

The boson field Ψ_a gives the form $\Psi_a dx^a$.

For spin $\frac{5}{2}$ none of the eqs. will do (one cannot construct the interaction)

— „ — $\frac{3}{2}$ it has just been done (supergravity)

Idea of I. M. G. possibly, it need not be $d\delta + \delta d$.

GELFAND-P000852

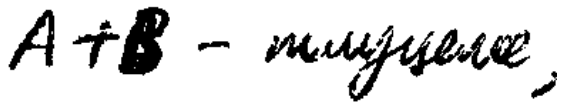
[source number: 3]

Equivalent form of the expression for Δ:

$$\Delta = -\Box + X, \tag{23}$$

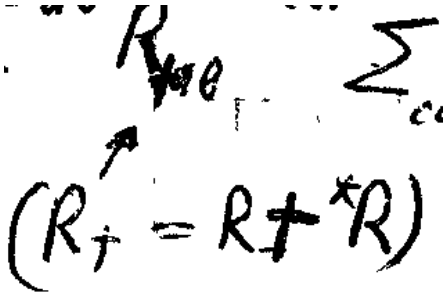
$$X = -\Sigma^{ab} R^{cd}_{ab} \Sigma_{cd}, \qquad A + B \text{ — integer} \tag{24}$$

$$X = -\frac{1}{A} \Sigma^{ab} R^{cd}_{+ab} \Sigma_{cd}, \qquad A + B \text{ — half-integer}, \quad A > B. \tag{25}$$



Original source details: GELFAND-P000852

Uncertain reading. The letter B in $A + B$ is heavily retraced. Its underlying form is indistinguishable.



Original source details: GELFAND-P000852

Diagram and typed description.

R^{cd}_{+ab}

$\uparrow$

$(R_+ = R + {}^*R)$

The arrow runs from the insertion to the overwritten lower sign at R in (25).

φ is symmetric here in $\alpha_1 \ldots \alpha_{2A}$ and in $\dot\alpha_1 \ldots \dot\alpha_{2B}$.

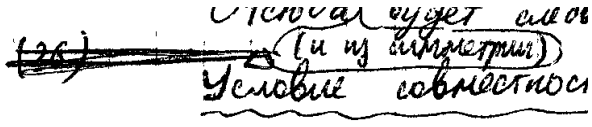
$$\Delta\varphi = \lambda\varphi \tag{26}$$



Original source details: GELFAND-P000852

Uncertain reading. The sign after Δ in (26) is overwritten. It is provisionally set as φ.

The compatibility condition will follow from this.



Original source details: GELFAND-P000852

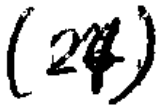
Diagram and typed description.

~~(26)~~ → (and from symmetry)

The stroke through the crossed-out number (26) becomes an arrow to the circled insertion.

Compatibility condition:

$$\nabla^{(\alpha_1}_{\dot\alpha} \nabla^{\alpha_2)\dot\alpha} \varphi_{\alpha_1\alpha_2\ldots\alpha_{2A},\dot\alpha_1\ldots\dot\alpha_{2B}} = 0 \tag{27}$$



Original source details: GELFAND-P000852

Uncertain reading. The last digit is overwritten. The label is provisionally set as (27). (28) is possible.

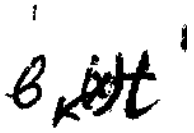
$A \le 1$, $B \le 1 \Rightarrow$ the case of all phys. fields

Consider the heat eq.

$$\frac{d}{dt}F(x,x',t) + \Delta F(x,x',t) = 0, \qquad F\big|_{t=0} = \delta(x-x'). \tag{28}$$

on a compact man.

$$\operatorname{Tr} F(x,x,t) \underset{t\to+0}{\sim} \sum_{k=0}^{\infty} b_k\, t^{(k-4)/2} \tag{29}$$



Uncertain reading. There is an overwritten mark before t, possibly a crossed-out (x).

Original source details: GELFAND-P000852



Undeciphered passage. The earlier writing before the exponent $(k-4)/2$ is crossed out and illegible.

Original source details: GELFAND-P000852

$$B_k = \int d^4x\sqrt{g}\, b_k, \tag{30}$$

Then

$$B_4 = n + m. \tag{31}$$

[At right:] B_4 — the first nondivergent one.

n — number of zero eigenval. (26)

m — [crossed out: ~~number of nonzero eigenval.~~] [Inserted below:] multiplicity of a nonzero (any) eigenval.

[At right:] [their common multiplicity — all multiplicities are the same!!]

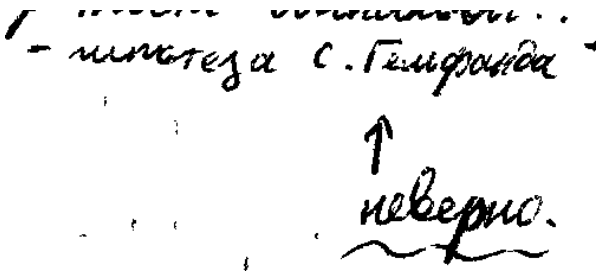


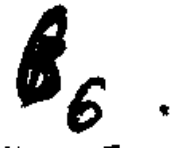
Diagram and typed description.
— S. Gelfand's conjecture
↑
incorrect.

An arrow from the underlined "incorrect" points upward to the conjecture.

Original source details: GELFAND-P000852

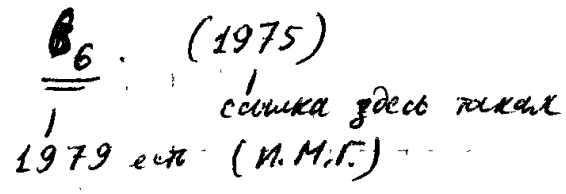
Aim — expr. in terms of top. invariants.

Gilkey got as far as B_6. (1975)



Uncertain reading. The overwritten letter is provisionally set as B. b is possible.

Original source details: GELFAND-P000852



Original source details: GELFAND-P000852

Diagram and typed description.

B_6 (1975)

the reference here is this one

1979 exists (I. M. G.)

Two short strokes link the upper symbols to the notes beneath them.



Original source details: GELFAND-P000852

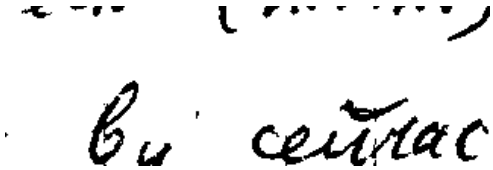
Uncertain reading. The word after 1979 is provisionally read «есть» (“exists”). «ещё» (“also”) is possible.



Original source details: GELFAND-P000852

Uncertain reading. The last word after “the reference here” is provisionally read «такая» (“this one”).

If there is a boundary, then likewise b_4 now [crossed out: ...] recently written up; the question is consid. difficult.



Original source details: GELFAND-P000852

Uncertain reading. The coefficient on the line about the boundary is provisionally read as b_4.



Original source details: GELFAND-P000852

Undeciphered passage. The word before “recently” is crossed out and illegible.

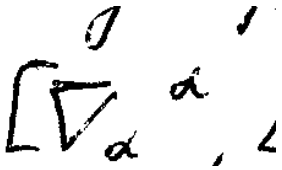
Consid. reducible reprs.

$$\varphi_{\alpha\beta} = \varphi_{(\alpha\beta)} + \frac{1}{2}\varepsilon_{\alpha\beta}\varphi^{\alpha}_{\alpha} \tag{32}$$

$$[1,0] = (1,0) \oplus (0,0)$$

By the compatibility conditions

$$[\nabla^{\dot\alpha}_{\alpha}, \Delta]\varphi[A,B] = 0 \tag{33}$$



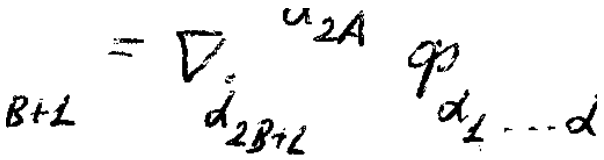
Original source details: GELFAND-P000852

Uncertain reading. There is a small stroke above the upper α. It is provisionally taken as a dot.

$$(\Delta - \lambda)\varphi_{\alpha_1 \ldots} = 0 \tag{34}$$

Consid.

$$\Psi_{\alpha_1 \ldots \alpha_{2A-1} \dot\alpha_1 \ldots \dot\alpha_{2B+1}} = \nabla^{\alpha_{2A}}_{\dot\alpha_{2B+1}} \varphi_{\alpha_1 \ldots \alpha_{2A} \dot\alpha_1 \ldots \dot\alpha_{2B}} \tag{35}$$



Uncertain reading. In (35) the dot belongs to the lower derivative index. The upper index is provisionally read as α_{2A}.

Original source details: GELFAND-P000853

$\Psi \neq 0$ when $\varphi \neq 0$ (otherwise $\overset{\circ}{\times}$ with (34)).



Uncertain reading. The small oval above the diagonal cross is provisionally set as a circle. The meaning of the mark is unclear.

Original source details: GELFAND-P000853

$$m[A, B] = m[A - \tfrac{1}{2}, B + \tfrac{1}{2}] \tag{36}$$

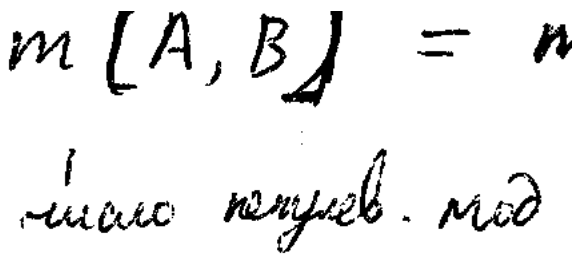


Diagram and typed description.
$m[A, B]$
|
number of nonzero modes
A short stroke below m points to the label.

Original source details: GELFAND-P000853

$$A + B = S, \qquad A - B = T.$$

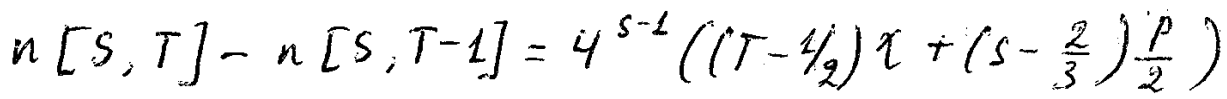
Then we get

$$m[S, T] = m[S] \tag{37}$$

Hence

$$n[S, T] - n[S, T'] = B_4[S, T] - B_4[S, T'] \tag{38}$$

$$n[S, T] - n[S, T - 1] = 4^{s-1}\big((T - \tfrac{1}{2})\chi + (s - \tfrac{2}{3})\tfrac{p}{2}\big) \tag{39}$$

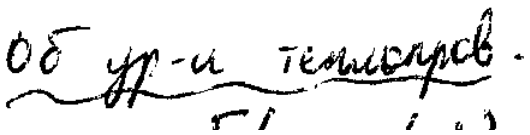


Original source details: GELFAND-P000853

Uncertain reading. The two letters at right in (39) are provisionally read as lowercase s. The notation at left looks like uppercase S.

χ — Euler number, p — Pontryagin number

On the heat eq.



Original source details: GELFAND-P000853

Uncertain reading. The end of the underlined heading is provisionally read «теплопров.» (“heat”).

$$F(x,x',t)=\sum_{i=0}^{\infty}\varphi_i(x)\varphi_i(x')e^{-\lambda_i t} \tag{40}$$

φ_i — orthonorm. syst.

$$F(x,x',0)=\delta(x,x') \tag{41}$$

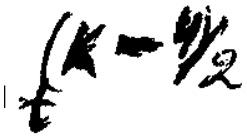
$$\int_M d^4x\sqrt{g}\,F(x,x,t)=\sum_{i=0}^{\infty}e^{-\lambda_i t}=n+\sum_{i=n+1}^{\infty}e^{-\lambda_i t} \tag{42}$$



Original source details: GELFAND-P000853

Uncertain reading. There is a small blacked-out mark above the second x, possibly a prime.

$$F(x,x,t)\sim\sum_{k=0}^{\infty}b_k(x)t^{(k-4/2} \tag{43}$$



Original source details: GELFAND-P000853

Uncertain reading. The 4 is overwritten. A separate closing parenthesis before the fraction slash cannot be distinguished.

m is determ. from this!! — by discarding the diverg. terms of $\sum e^{-\lambda_i t}$ and n contributes.

m — regulariz. sum

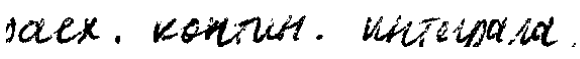


Original source details: GELFAND-P000853

Uncertain reading. The abbreviation in the underlined note is provisionally read «регуляриз.» (“regulariz.”).

“m — is an integer (!) on deep top. grounds ”

These are the stand. one-loop diverg. of the cont. integral.



Original source details: GELFAND-P000853

Uncertain reading. The Russian end of the line is provisionally read «контин. интеграла». The last word looks like a genitive form.

[At right: в?]

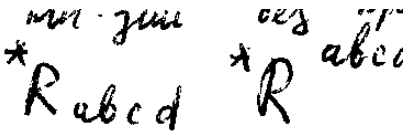


Uncertain reading. The isolated dark sign may be the letter «в». Its reading is uncertain.

Original source details: GELFAND-P000853

Def. χ and p for manifs. without boundary

$$\chi = \int {}^*R_{abcd}\, {}^*R^{abcd}\, d^4x\sqrt{g} \tag{44}$$



Uncertain reading. Stars are visible before both occurrences of R in (44).

Original source details: GELFAND-P000853

$$p = \int {}^*R_{abcd} R^{abcd}\, d^4x\, \sqrt{g} \tag{45}$$

GELFAND-P000854

[source number: 4]

8.10.79. Gelfand Sem.

Draft theor. minimum, needed in any specialization. (for 1st-year students)

1. Algebra

van der Waerden, chs. 1–5, 12.

Gelfand lectures on lin. algebra

Atiyah, Macdonald

2. Analysis

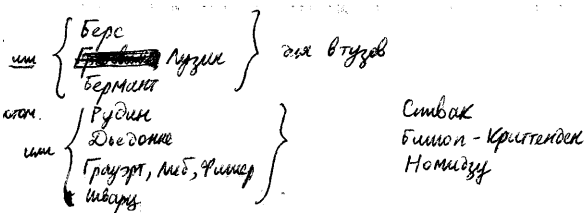


Diagram and typed description.

or { Bers / [crossed out: ...] Luzin / Berman } for technical institutes.

then.

or { Rudin / Dieudonné / Grauert, Lieb, Fischer / Schwartz }

[At right:]
Spivak
Bishop–Crittenden
Nomizu

Two pairs of braces group the lists. Three separate lines stand to the right of the second pair.

Original source details: GELFAND-P000854



Original source details: GELFAND-P000854

Uncertain reading. A surname is crossed out before “Luzin.” “Berman” is a possible reading.

Check question: calculate approximately

$$\int_{-\infty}^{\infty} e^{-x^2} \cos 1000x \, dx, \qquad \int_{-\infty}^{\infty} e^{-1000x^2} \cos x \, dx$$

(and comp. with tables)

$$\int_{-\infty}^{\infty} e^{-1000x^4} \cos x \, dx, \qquad \int_{-\infty}^{\infty} e^{-x^4} \cos 1000x \, dx$$

3. Arnold I, II, III.

4. Kirillov $\frac{1}{2} + \frac{1}{2}$

Serre lin. reprs. of finite groups

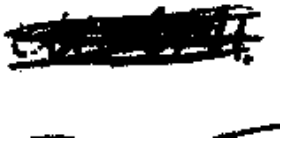
(he does not understand the 3rd chapter)

Generalized fns. I issue, chs. 1–2

5. Know the partial differential equations. (there is nothing to read).

Check problem in lin. algebra:

find the can. form of a self-adj. oper. in a [crossed out: ...] symplect. space.



Original source details: GELFAND-P000854

Undeciphered passage. The word after “in” is crossed out and illegible.



Original source details: GELFAND-P000854

Uncertain reading. The abbreviation after the crossed-out word is provisionally read «симплект.» (“symplect.”).

6. [blacked out] Spec. theory of rel. + GR



Original source details: GELFAND-P000854

Uncertain reading. A single sign before “Spec.” is blacked out. It may be the letter «А».

+ geometry

[At right:] Landau, Lifshitz (vol. 2)

jointly with Nomizu



Original source details: GELFAND-P000854

Uncertain reading. The abbreviation before “with Nomizu” is provisionally read «совм.» (“jointly”).

S. Gindikin. Beginnings of twistor theory.

To be presented in the language of integr. geometry.

[At right:] Twistors were introduced by Penrose.

see one of the recent issues of BAMS,

1979, vol. 1 (new series),

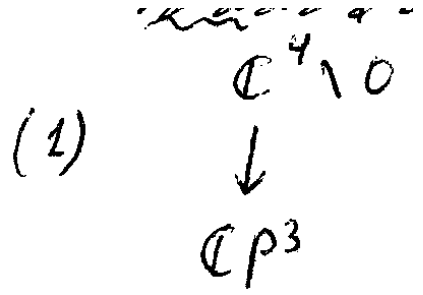
art. by Wells (Wells.)

[crossed out: ~~... let us consider?~~]



Original source details: GELFAND-P000854

Undeciphered passage. The crossed-out line before (1) cannot be read in full. The last word is provisionally read «рассмотрим» (“let us consider”).



Original source details: GELFAND-P000854

Diagram and typed description.

$$(1) \qquad \begin{array}{c} \mathbb{C}^4 \setminus 0 \\ \big\downarrow \\ \mathbb{C}P^3 \end{array}$$

In $\mathbb{C}^4$ fix a Hermitian form [blacked out] $(++--)$. $\langle\ ,\ \rangle$.



Original source details: GELFAND-P000854

Undeciphered passage. A sign is blacked out before the parentheses with the signature. It is illegible.

We obtain α-twistors: elements of $\mathbb{C}P^3$ or corresp. elems. of $\mathbb{C}^4$.

$(\mathbb{C}P^3)'$ — dual sp.. A form arises there too, and these are — β-twistors.

$$\langle z, z\rangle > 0 \Rightarrow \text{pos. twistor}$$
$$\langle z, z\rangle = 0 \Rightarrow \text{null} — „ —$$
$$\langle z, z\rangle < 0 \Rightarrow \text{neg.} — „ —$$

In $\mathbb{C}P^3$ consid. lines (2-dim. subsp. in $\mathbb{C}^4$). Their set: $G = G_{4,2} = G^{\mathbb{C}}_{4,2}$.

Over G there is a natural princ. bundle of 2-frames in $\mathbb{C}^4$:



Original source details: GELFAND-P000855

Uncertain reading. The word before “bundle” is provisionally read «натурельное» (“natural”). Its middle and final strokes run together.



Original source details: GELFAND-P000855

(2)

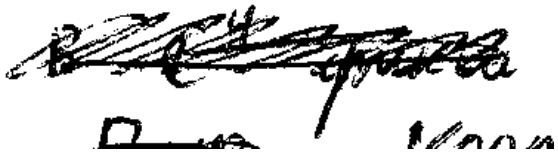
E

$\downarrow$

G

Diagram and typed description.

[Crossed out above the heading: ~~In $\mathbb{C}^4$ ...~~]



Original source details: GELFAND-P000855

Undeciphered passage. The crossed-out line may begin «В $\mathbb{C}^4$» (“In $\mathbb{C}^4$”). The full line is illegible.

[Crossed out at left: ~~...~~] Coordinates in G.



Original source details: GELFAND-P000855

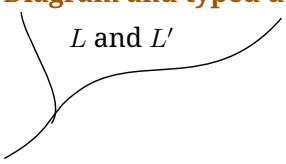
Undeciphered passage. A separate short entry at the left of the heading is crossed out and illegible.

Choose in $\mathbb{C}^4$ 2 planes (transvers.) L and L', then any other is a graph, the matrix of the corresp. transf. gives coordinates.



Original source details: GELFAND-P000855

Diagram and typed description.



The insertion mark leads to the plane labels above the line.

Choose in [crossed out: ...] L basis $e_0\ e_1$; In L' basis e_3, e_4.



Original source details: GELFAND-P000855

Undeciphered passage. The entry before L is crossed out and illegible.

Then if $\widetilde{L} \in G_{4,2}$, then in $\widetilde{L}$ there is a basis of the form



Original source details: GELFAND-P000855

Uncertain reading. The repeated plane symbol is heavily retraced. It is provisionally set as $\widetilde{L}$.

$$\begin{cases} f_0 = e_0 + u_{03}e_3 + u_{04}e_4 \\ f_1 = e_1 + u_{13}e_3 + u_{14}e_4 \end{cases} \tag{3}$$



Original source details: GELFAND-P000855

Uncertain reading. The first subscript of u before e_3 in the second line is provisionally read as 1. The numeral has been retraced.

u_{ij} — coordinates.

Let L, L' be transvers. and isotropic with resp. to $\langle \cdot, \cdot \rangle$. Problem. Such L and L' exist.

$$\{e_A,\ A = 0, 1\} \text{ — basis in } L. \tag{4}$$

In L' introduce a basis

$$\{e^{A'},\ A' = 0', 1'\} \text{ — basis in } L' \tag{5}$$

[crossed out: ...] [above the line: Let [crossed out: ...]] the form [above the line: $\langle\ ,\ \rangle$] in the basis $(e_0, e_1, e^{0'}, e^{1'})$ have [crossed out: ...] the matrix



Original source details: GELFAND-P000855

Undeciphered passage. The entry before "the form" on the main line is crossed out and illegible.



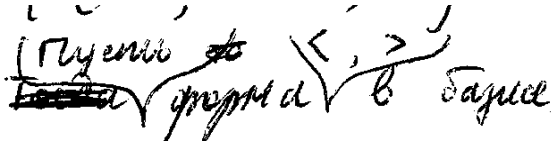
Original source details: GELFAND-P000855

Undeciphered passage. A short mark after the inserted "Let" is crossed out and illegible.



Original source details: GELFAND-P000855

Undeciphered passage. The word between “have” and “the matrix” is crossed out and illegible.



Original source details: GELFAND-P000855

Diagram and typed description.

Let ... $\langle , \rangle$

... the form in the basis

Two marks give the insertion points for “Let” and the angle brackets.

$$\begin{pmatrix} 0 & iE \\ -iE & 0 \end{pmatrix}; \tag{6}$$

i.e.

$$\langle e_0, e^{0'} \rangle = \langle e_1, e^{1'} \rangle = i, \qquad \text{the other scal. prod.} = 0 \tag{7}$$

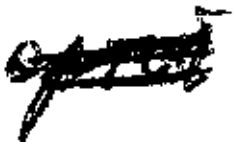
Make a change of variables in L:

$$\widetilde{e}_B = g_B^A e_A \tag{8}$$

Then it is easy to check that

$$e^{A'} \text{ transf. by means of } (g^*)^{-1} \tag{9}$$

(follows from the relat. of [crossed out: ...] orthogonality)



Original source details: GELFAND-P000855

Undeciphered passage. The entry before “orthogonality” is crossed out and illegible.

There are other transf. laws: [Crossed out at left: ...]



Original source details: GELFAND-P000855

Undeciphered passage. The mark before the lines with indices is crossed out and illegible.



Original source details: GELFAND-P000855

Diagram and typed description.

A' by $\bar{g}$ $\qquad$ B' by $(g^*)^{-1}$

B' — „ — $(g^T)^{-1}$

The arc is above A' and below B and B'. The meaning of the arcs is unclear.

GELFAND-P000856

[source number: 5]

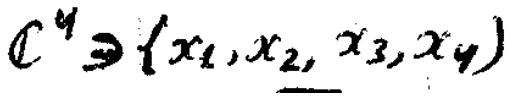
Problem. Prove that $U(2,2)$ has a subgroup isom. to $GL(2,\mathbb{C})$ (hint: use the preceding result).

Coordinates:

$$\{z^A, z_{A'}\} \qquad \text{in the basis } \{e_A, e^{A'}\} \tag{10}$$

$$z = z^A e_A + z_{A'} e^{A'} \in \mathbb{C}^4 \tag{11}$$

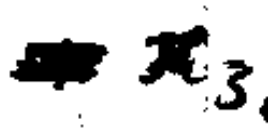
I.M.G. $\mathbb{C}^4 \ni \{x_1, x_2, x_3, x_4\}$.



Original source details: GELFAND-P000856

Uncertain reading. The opening mark resembles a brace. The closing mark is a parenthesis.

$$\langle x, y\rangle = x_1\bar{y}_1 + x_2\bar{y}_2 - x_3\bar{y}_3 - x_4\bar{y}_4 \tag{12}$$



Original source details: GELFAND-P000856

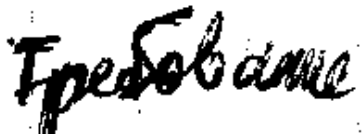
Uncertain reading. The mark before the third term is provisionally read as a minus sign. The stroke has been overwritten.

L, L' — 2 isotropic subsp. and 2 bases $\{e_0, e_1\}$; $\{e^{0'}, e^{1'}\}$. Then

$$z = z^0 e_0 + z^1 e_1 + z_{0'} e^{0'} + z_{1'} e^{1'} \tag{13}$$

$$\{z^A, z_{A'}\} \text{ — spinor coordinates in } \mathbb{C}^4 \tag{14}$$

Requirement on the basis:



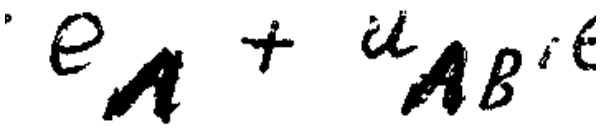
Original source details: GELFAND-P000856

Uncertain reading. The initial letter of «Требование» (“Requirement”) is retraced.

$$\langle e_0, e^{0'}\rangle = \langle e_1, e^{1'}\rangle = i, \qquad \text{the other scal. prod.} = 0. \tag{15}$$

S.G. Let $\widetilde{L} \in G$; $\widetilde{L} \cap L' = \{0\}$. Then choose a basis as follows:

$$f_A = e_A + u_{AB'} e^{B'} \tag{16}$$



Original source details: GELFAND-P000856

Uncertain reading. The two indices on the left and in the first term on the right are retraced and read as A.

The spinor indices themselves show how they transform. (worth writing out!)

(Twistor theory in curved space has yet to be worked out, though there are already papers

on it). Consider groups acting in $\mathbb{C}^4$. The action of $GL(4,\mathbb{C})$ carries over to $G_{4,2}$.

$$\begin{aligned} T_+ &: \quad \langle z,z\rangle > 0 \\ T_- &: \quad \langle z,z\rangle < 0 \\ N &: \quad \langle z,z\rangle = 0 \qquad \text{— 7-dimensional surface in } \mathbb{C}^4. \end{aligned}$$

S_+ — 2-dimensional planes $\subset T_+$ (except 0)
S_- — — „ — $\subset T_-$ (— „ —)
M — — „ — $\subset N$

S_+, S_- — lie entirely within the range of the coord. $u_{AB'}$.

$$\begin{cases} S_+ = \{u : \ \frac{1}{i}(u-u^*) \gg 0\} \\ S_- = \{u : \ \frac{1}{i}(u-u^*) \ll 0\} \end{cases} \tag{17}$$

GELFAND-P000857

M no longer lies wholly in the coord. neighborhood. Let M_0 — be the part that does lie in it.

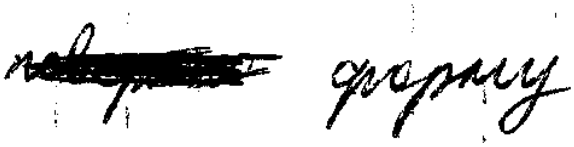
$$M_0 = \{u : \ u = u^*\} \tag{18}$$

$$\dim_{\mathbb{R}} M_0 = 4, \qquad \dim_{\mathbb{C}} S^{\pm} = 4.$$

M_0 — the common skeleton of $S^{\pm}$.

$$\partial S_+ = \{\text{supporting planes to } T_+\}$$

Now consider in $\mathbb{C}^4$ the group $U(2,2)$, preserving [crossed out: ...] the form



Undeciphered passage. The entry before “the form” is crossed out and illegible.

Original source details: GELFAND-P000857

(and, hence, the surface N). Then it is clear that $U(2,2)$ preserves all [crossed out: ...] the domains $S_\pm, M$



Undeciphered passage. The entry before “domains” is crossed out and illegible.

Original source details: GELFAND-P000857

(canon. construction) and acts on each transitively!! (A problem in lin. algebra!)

On $(\mathbb{C}^4, \langle\ ,\ \rangle)$ acts $U(2,2)$ and preserves the whole construction. $g \in U(2,2)$ is written in a basis of the kind descr. above:

$$g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{19}$$

Action on u:

$$g:\ u \mapsto (Au + B)(Cu + D)^{-1} \tag{20}$$

The action makes M into the conformal compactification of Minkowski space: the Poincaré gr. $\subset SU(2,2)$ (addition of conformal transf.).

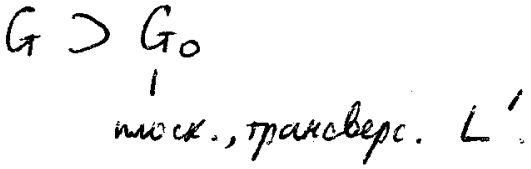


Diagram and typed description.

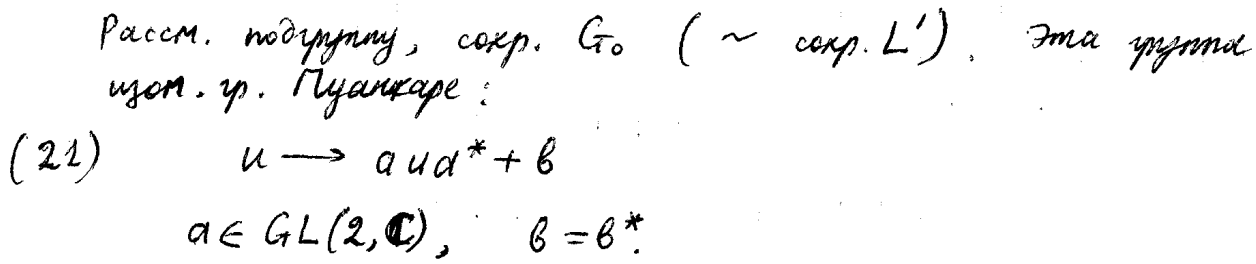
$G \supset G_0$
|
planes transv. to L'.

Original source details: GELFAND-P000857

Consider the subgroup preserving G_0 ($\sim$ preserving L'). This group is isom. to the Poincaré gr.:

$$u \to aua^* + b \tag{21}$$

$a \in GL(2,\mathbb{C})$, $b = b^*$.



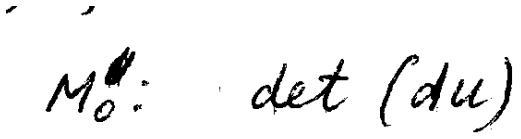
Original source details: GELFAND-P000857

Manuscript excerpt. The manuscript reads “This group is isom. to the Poincaré gr.” Equation (21) and its parameters follow.

In the notation of (19) here:

$$A = a, \quad C = 0, \quad D = (a^*)^{-1}, \quad BD^{-1} = b. \tag{22}$$

The Minkowski metric on $M_0^{\cdots}$: $\det(du)$ (23)

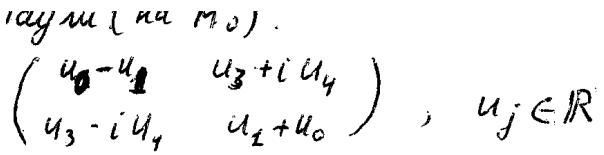


Uncertain reading. The stroke above M_0 is obscured. The superscript is illegible.

Original source details: GELFAND-P000857

Pauli coord. (on M_0):

$$\begin{pmatrix} u_0 - u_1 & u_3 + iu_4 \\ u_3 - iu_4 & u_1 + u_0 \end{pmatrix}, \qquad u_j \in \mathbb{R} \tag{24}$$



Uncertain reading. The two indices in the upper left entry are retraced and read as 0 and 1.

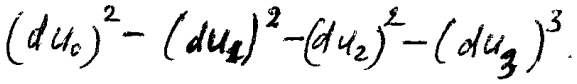
Original source details: GELFAND-P000857

GELFAND-P000858

[source number: 6]

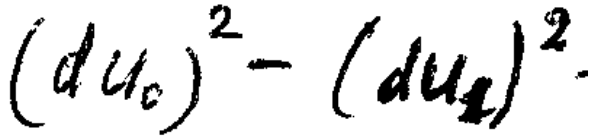
Then the metric takes the form

$$(du_0)^2 - (du_1)^2 - (du_2)^2 - (du_3)^3. \tag{25}$$



Uncertain reading. The last term has exponent 3 in the manuscript.

Original source details: GELFAND-P000858



Uncertain reading. The first term reads 0. The retraced subscript in the second term reads 1.

Original source details: GELFAND-P000858

$$M_0 = M \cap G_0 \underset{\text{def}}{=} \text{ordinary Minkowski space.} \tag{26}$$

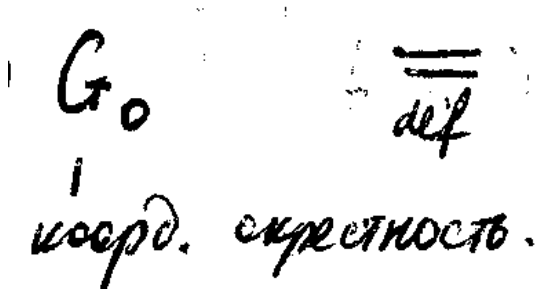


Diagram and typed description.

G_0

|

coord. neighborhood.

Original source details: GELFAND-P000858

A point of M — is a point of G or a line in $\mathbb{C}P^3$. Restrict attention to M_0.

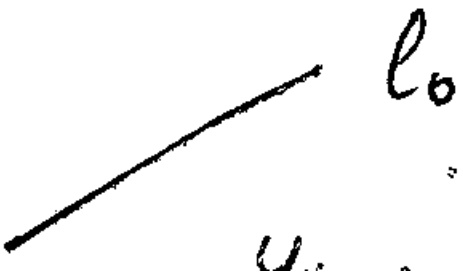


Diagram and typed description.

ℓ_0

Original source details: GELFAND-P000858

Consider all lines intersecting ℓ_0. Their eq. in coordinates:

$$\det(u^0 - u) = 0. \tag{27}$$

It turns out that this is [crossed out: ...] the Minkowski isotropy cone (the cone of null rays).



Original source details: GELFAND-P000858

Undeciphered passage. The word before "the Minkowski isotropy cone" is crossed out and illegible.

$$(27) \cap M = \{\text{light cone}\}.$$

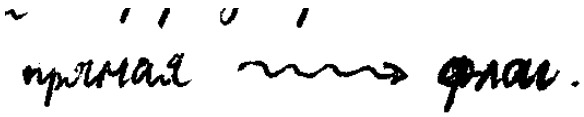
There is a one-to-one corresp. between points of M and rays on N.



Original source details: GELFAND-P000858

Diagram and typed description.
Light rays on M
$=$ isotropic points in $\mathbb{C}P^3$

Contest: formulate a higher-dimensional generalization of this situation.



Original source details: GELFAND-P000858

Diagram and typed description.
line ⇝ flag.

— · —

GELFAND-P000859

15.10.79. Gelfand Sem.

F. A. Berezin. Supermathematics.

$\mathbb{K}$ — a field ($\mathbb{R}$ or $\mathbb{C}$).

$\mathbb{Z}_2$-graded space $L = L_0 \oplus L_1$.

The elements of L_0 and L_1 are called homogeneous.

Parity

$$\alpha(x) = \begin{cases} 0, & x \in L_0 \\ 1, & x \in L_1. \end{cases}$$

Sums and prod. are taken mod2.

Grassm. algebra Λ over $\mathbb{K}$ — assoc. alg. with generators

$$1\ (\in \mathbb{K}), \quad \xi_1, \ldots, \xi_N$$

and relat.

$$\xi_i \xi_j + \xi_j \xi_i = 0, \qquad \forall i, j.$$

$$\{1,\ \xi_{i_1} \ldots \xi_{i_k},\ i_1 < i_2 < \ldots < i_k\} \quad \text{— a basis in } \Lambda \text{ over } \mathbb{K}.$$

$\Lambda = \Lambda_0 \oplus \Lambda_1$, Λ_0 is gen. by 1 and prod. of an even number of ξ_i.

GELFAND-P000860

[source number: 10]

$$\dim_{\mathbb{K}} \Lambda = 2^N.$$

For homogeneous f, g

$$f \cdot g = (-1)^{\alpha(f)\cdot\alpha(g)} g f. \tag{1}$$

Grassmann envelope of the lin. space L:

[crossed out: ~~(2) $L(\Lambda) = L \otimes_{\mathbb{K}} \Lambda$~~]



Original source details: GELFAND-P000860

Uncertain reading. The hatched-out entry provisionally reads (2) $L(\Lambda) = L \otimes_{\mathbb{K}} \Lambda$.

e_i — a basis in L of homog. elements ($L = L_0 \oplus L_1$).

Now consider

$$\left\{ \sum a_i e_i,\ a_i \in \Lambda,\ \alpha(a_i) = \alpha(e_i) \right\} \underset{\text{def}}{=} L(\Lambda). \tag{2}$$

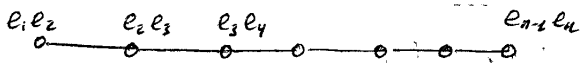
This is a left Λ_0-module. Equivalently:

$$L(\Lambda) = \left(\Lambda \underset{\mathbb{K}}{\otimes} L \right)_{\bar{0}}. \tag{2}'$$

Linear operators.

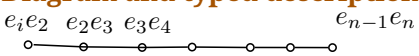
I.M.G. These alg. arise in the theory of quiver repr. (cf. FA, 1979; $N = 3$).

Conj.: $\forall$ Dynkin diagram there should be an analogue of the even part of the Grassm. algebra. (so that the ordinary one corresp. to the diagram A_n



Original source details: GELFAND-P000860

Diagram and typed description.



The dotted first subscript is read as i. No arrowheads can be made out on the chain.

if joined by an arrow, then the prod. is $= 0$, if unjoined, then there is a new element. Idea: there are other alg. of Grassmann type.

— · —

operator in L:



Original source details: GELFAND-P000860

Uncertain reading. There is a dark stroke at the right end of L. A separate subscript cannot be established.

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}. \tag{3}$$

Grading:

[crossed out: ...]



Undeciphered passage. The sign before the first matrix is crossed out. It may be A.

Original source details: GELFAND-P000860

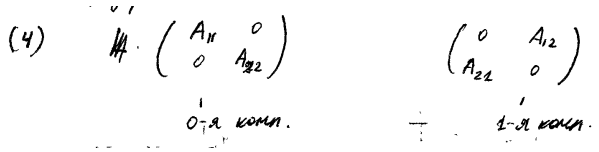
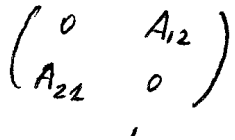


Diagram and typed description.

(4) $\begin{pmatrix} A_{11} & 0 \\ 0 & A_{22} \end{pmatrix}$ $\begin{pmatrix} 0 & A_{12} \\ A_{21} & 0 \end{pmatrix}$

0th comp. 1st comp.

Original source details: GELFAND-P000860



Uncertain reading. The lower left block in the second matrix is read as A_{21}.

Original source details: GELFAND-P000860

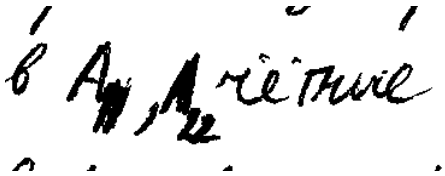
The space $L_{p,q}$ — all such oper., $p = \dim L_0$, $q = \dim L_1$.

$L_{p,q}(\Lambda)$ — Grassm. envelope:

Remark: a_i and e_i are always taken to commute.

$L_{p,q}(\Lambda)$ — also matrices of Grassm. numbers:

in A_{11}, A_{22} even



Uncertain reading. The first pair of subscripts is heavily retraced and provisionally reads 11. The second reads 22.

Original source details: GELFAND-P000860

in A_{12}, A_{21} odd.

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \in L_{p,q}(\Lambda).$$

$L_{p,q}(\Lambda)$ — is an algebra (and a Λ_0-module).

Supertrace

$$\operatorname{str} A = \operatorname{tr} A_{11} - \operatorname{tr} A_{22}. \tag{5}$$

Prop.:

$$\operatorname{str}(AB) = \operatorname{str}(BA) \tag{6}$$

Superdeterminant. (Berezinian):

$$\operatorname{sdet} = \det(A_{11} - A_{12}A_{22}^{-1}A_{21})(\det A_{22})^{-1} \tag{7}$$

(for numerical matrices there is a fmla. [crossed out: ...] of the same form, but with $\det A_{22}$ in place of $(\det A_{22})^{-1}$).

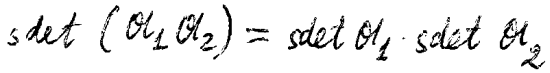
Undeciphered passage. The entry after "fmla." is crossed out and illegible.



Original source details: GELFAND-P000861

$$\operatorname{sdet}(\alpha_1\alpha_2) = \operatorname{sdet}\alpha_1 \cdot \operatorname{sdet}\alpha_2 \tag{8}$$

Uncertain reading. The symbols in (8) are provisionally set as α_1, α_2. They could be cursive capitals A_1, A_2.



Original source details: GELFAND-P000861

Exercise. Do this if $\dim L_0 = \dim L_1 = 1$, i.e. for matrices of order 2.

Explanation. Consider matrices with anticomm. entries:

$$\operatorname{sdet}\begin{pmatrix} 1 & \xi_1 \\ \xi_2 & 1 \end{pmatrix} = 1 - \xi_1\xi_2.$$

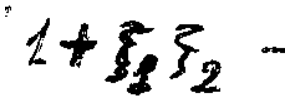
$$\begin{pmatrix} 1 & \xi_1 \\ \xi_2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1+\xi_1\xi_2 & 2\xi_1 \\ 2\xi_2 & 1+\xi_2\xi_1 \end{pmatrix}.$$

Diagram and typed description. ### A small hash-like sign sits to the right of the matrix. Its purpose is unclear.



Original source details: GELFAND-P000861

$$\text{sdet} = (1 + \xi_1\xi_2 - 2\xi_1(1 + \xi_1\xi_2)2\xi_2)(1 + \xi_1\xi_2) =$$
$$= (1 - 3\xi_1\xi_2)(1 + \xi_1\xi_2) = 1 - 2\xi_1\xi_2 = (1 - \xi_1\xi_2)^2.$$



Uncertain reading. In the first product after $1+$, the first subscript is retraced and read as 1.

Original source details: GELFAND-P000861

Reduction of matrices to canon. form.

$L(\Lambda)$

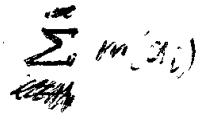


Uncertain reading. The separate faded entry is provisionally read as $L(\Lambda)$.

Original source details: GELFAND-P000861

There is a mapping

$$m. \quad L(\Lambda) \longrightarrow L_0$$
$$\sum a_i e_i \longmapsto \left(\sum a_i e_i\right)\Big|_{\xi=0} = \sum_i m(a_i)e_i,$$
$$m(a_i) = a_i|_{\xi=0}.$$



Undeciphered passage. The lower entry at the last summation is crossed out. An undeciphered stroke is visible above the sign.

Original source details: GELFAND-P000861

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \in L_{p,q}(\Lambda).$$



Uncertain reading. The stroke of L in the second matrix line is broken. The expression is provisionally read as $L_{p,q}(\Lambda)$.

Original source details: GELFAND-P000861

$$m(A) = \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix}. \tag{9}$$

Let $\mathbb{K} = \mathbb{C}$.

Theorem.

Let $m(A)$ have distinct eigenv.. Then A also reduces to diag. form:

$$BAB^{-1} = \begin{pmatrix} c_1 & & 0 \\ & \ddots & \\ 0 & & c_{p+q} \end{pmatrix}. \tag{10}$$

GELFAND-P000862

[source number: 11]

Analogues of Jordan blocks are unknown here.

Problem 1. Do this (analogue of normal forms)!

A problem related to instantons:

Problem 2. Consid. the matrix $A = (a_{ik})_{i,k=1}^{n}$ with entries $a_{ik} \in \Lambda_1$ (odd Grassm. numbers).

Describe all $B = (b_{ik})_{i,k=1}^{n}$, with entr. $b_{ik} \in \Lambda_0$ such that

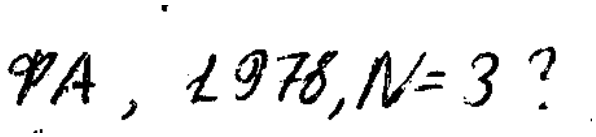
$$AB = BA.$$

If Λ has 4 generators, then it has a relation to Y.-M. If all entr. of A are independent (in Λ), then it can be solved.

— · —

S.G. Connection with the work of Beilinson and Bernst.–Gelf.–Gelf.

(see FA, 1978, $N = 3$?; or Drinfeld, Manin; Nucl. Physics, $N = 6$, 1979: the constr. is given explicitly there, translated into the language of solutions).

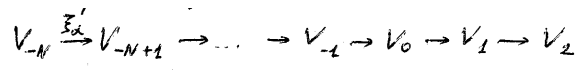


Original source details: GELFAND-P000862

Uncertain reading. The question mark after $N = 3$ is in the manuscript.

classif. of vect. bundles on $\mathbb{C}P^3$. (holom.)

Objects — graded modules over $\Lambda = \Lambda(4)$:



Original source details: GELFAND-P000862

Diagram and typed description.

$V_{-N} \xrightarrow{\xi_\alpha} V_{-N+1} \longrightarrow \cdots \longrightarrow V_{-1} \to V_0 \to V_1 \to V_2$

$\alpha = 1, 2, 3, 4$. Repr. of the Grassm. algebra:

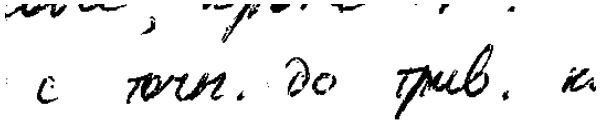
$$\xi_i \xi_j + \xi_j \xi_i = 0.$$

$a_1\xi_1 + a_2\xi_2 + a_3\xi_3 + a_4\xi_4$ — an operator from V_k to V_{k+1}

and we get a complex;

the cohom. vanish except for H^0.

Such modules up to triv. classify instantons.



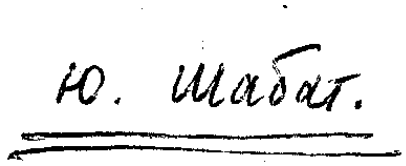
Original source details: GELFAND-P000862

Uncertain reading. The abbreviation after “up to” provisionally reads «трив.» (“triv.”).

Problem: give an explicit construction. It has yet to be done (the spaces are constructed, but the mappings are missing).

Problem: give the classif. for the Grassmannian!!

Yu. Shabat.



Original source details: GELFAND-P000862

Uncertain reading. The manuscript gives the initial «Ю.» (“Yu.”). MCCME catalogue record 469 gives the name as “G.B. Shabat”.

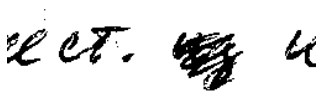
Lie–Wirtinger theorem (classical)

$\Gamma \subset \mathbb{C}^n$ — a germ of an analytic hypersurface. It is called a double translation surf., if it can be represented in 2 ways as a sum

$$\Gamma = \gamma_1 + \ldots + \gamma_{n-1}, \qquad \Gamma = \delta_1 + \ldots + \delta_{n-1}$$

γ_i, δ_i — germs of analytic curves.

$\{\gamma_i\}$ and $\{\delta_i\}$ cannot be obtained from one another by perm. [crossed out: ...] and translations.



Original source details: GELFAND-P000862

Undeciphered passage. The entry after “perm.” is crossed out and illegible.

Example: X — a smooth curve of genus n.

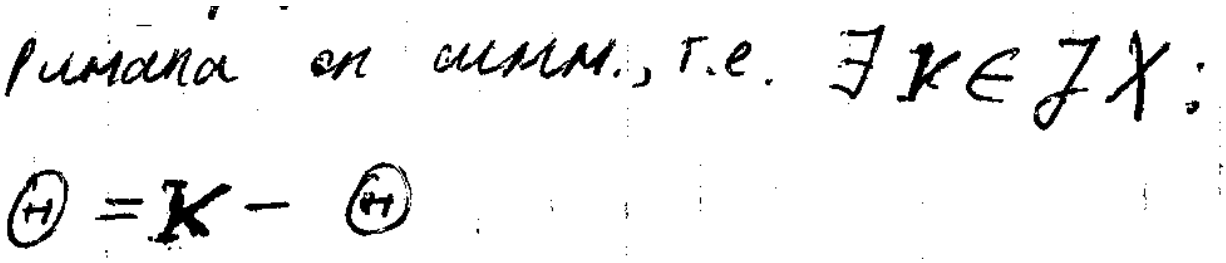
JX — the Jacobian of the curve X; $\quad j : X \longrightarrow JX$.

GELFAND-P000863

$$\Theta = \underbrace{j(X) + \ldots + j(X)}_{n-1 \text{ times}}, \qquad (\Theta \text{ — Poincaré divisor}).$$

By Riemann's thm. it is symm., i.e. $\exists K \in JX$:

$$\Theta = K - \Theta.$$



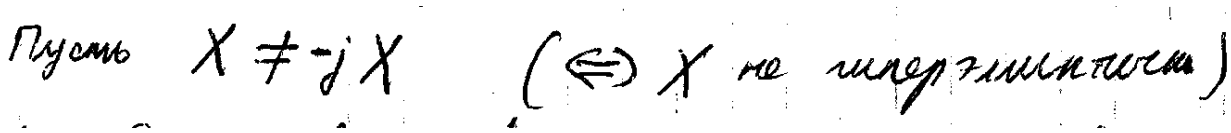
Original source details: GELFAND-P000863

Uncertain reading. The sign is set as K in both places. Another possible reading is κ.

Therefore up to translation

$$\Theta \sim \underbrace{-jX - \ldots - jX}_{n-1}.$$

Let $X \neq -jX$ ($\Leftrightarrow X$ is nonhyperelliptic).



Original source details: GELFAND-P000863

Uncertain reading. The manuscript has X on the left with no preceding j.

Assertion: all translation surf. are these surf. and their degenerations
(if Γ — a double translation surf., then a small deform. of Γ — a germ of $\Theta \subset JX$).
Conjecture of I.M.G.: 1) All integr. systems have this theorem as their basis.
2) This thm. is the first in a series of coloss. theorems of this kind.



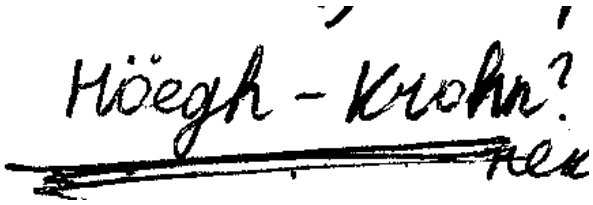
Uncertain reading. The abbreviation «колосал.» is a provisional reading. The middle strokes run together.

Original source details: GELFAND-P000863

1) — the reason for the appearance of Θ-funcs..

— · —

Höegh-Krohn? someone from Norway. (interpreted by Gelfand).



Uncertain reading. The manuscript puts a question mark after the name. MCCME catalogue record 470 gives “Raphael Hoegh-Krohn”.

Original source details: GELFAND-P000863

On mapping grps.
(Ismagilov, Graev, Vershik; Albeverio)
[below the list: and ...]
X — a Riem. manifold
G — a compact Lie gr. (simple)

$$\varphi \in C_0^\infty(X, G) \equiv G^X.$$



Uncertain reading. The open-loop sign is provisionally set as φ. Another possible reading is ψ.

Original source details: GELFAND-P000863

We study unit. representations.

g — the Lie alg. of G.

[crossed out at left: ...] $\Omega(X, g)$ — 1-forms with values in the Lie alg..

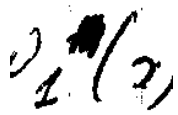


Undeciphered passage. The sign before $\Omega(X, g)$ is crossed out and illegible.

Original source details: GELFAND-P000863

$$\omega_i \in \Omega(X, g)$$

$$(\omega_1, \omega_2) = \int_X \operatorname{tr}\big(\omega_1(x)\omega_2^*(x)\big)\, dx. \tag{1}$$



Uncertain reading. The dark stroke above ω_1 may be a crossed-out star. Its reading is uncertain.

Original source details: GELFAND-P000863

$$\omega_i(x) : T_x \longrightarrow g$$

there is an inner prod. in both spaces (on the right, Killing)

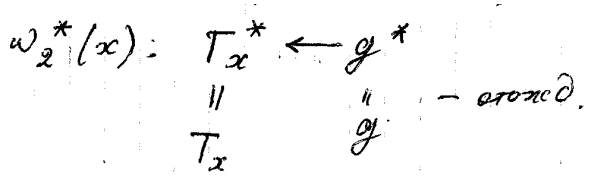


Diagram and typed description.

$$\omega_2^*(x): \quad T_x^* \longleftarrow g^* \qquad \text{— identif.}$$
$$\qquad\quad \| \qquad\quad \| \qquad\quad \text{— atomic.}$$
$$\qquad\quad T_x \qquad\quad g$$

Original source details: GELFAND-P000863

GELFAND-P000864

[source number: 12]

$$\omega_1\omega_2^* : g \longrightarrow g$$

the trace and the integr. can now be taken

$\mathcal{H}$ — a Hilbert sp., completion of $\Omega(X, g)$

$$L_2(\mathcal{H}, \mu)$$

$$\dim \mathcal{H} < \infty \quad \Rightarrow \quad d\mu \sim e^{-\frac{1}{2}|x|^2}\, dx$$



Uncertain reading. The ∞ sign is retraced. A separate stroke is visible beneath it.

Original source details: GELFAND-P000864

(and analog. in an [crossed out: ~~finite~~] ∞-dim. sp.)



Uncertain reading. The crossed-out word provisionally reads «конечно» (“finite”).

Original source details: GELFAND-P000864

$T \in E(\mathcal{H})$ — Euclidean transf. (motions) of the space $\mathcal{H}$,

$$Tx = Ox + a, \qquad x \in H, \quad a \in H$$

μ is inv. under O and quasi-inv. under translations.

$g \in E(\mathcal{H})$, $g \to U_g$ unitary on $L_2(\mathcal{H}, \mu)$

hence [«ср.»] a unit. repr. of $E(\mathcal{H})$ in $L_2(\mathcal{H}, \mu)$



Uncertain reading. The abbreviation is provisionally read «ср.». The second sign has a long descending stroke.

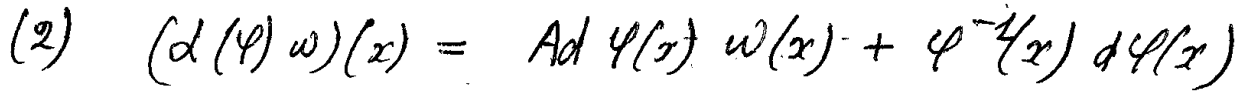
Original source details: GELFAND-P000864

$$(\alpha(\varphi)\omega)(x) = \mathrm{Ad}\,\varphi(x)\,\omega(x) + \varphi^{-1}(x)\,d\varphi(x). \tag{2}$$



Uncertain reading. The open-loop sign is provisionally set as φ. Another possible reading is ψ.

Original source details: GELFAND-P000864



Original source details: GELFAND-P000864

Uncertain reading. The manuscript shows the order $\varphi^{-1}(x)\,d\varphi(x)$ and the exponent -1.

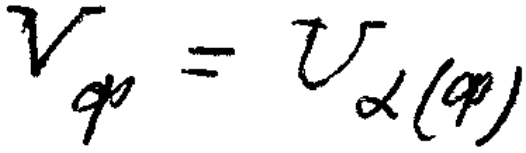
$$\alpha : C_0^\infty(X, G) \longrightarrow E(\Omega(X))$$

— this is a group homomorphism:

$$\alpha(\varphi_1\varphi_2) = \alpha(\varphi_1)\alpha(\varphi_2).$$

Take the comp. with U_g;

$\underline{\underline{V_\phi = U_{\alpha(\phi)}}}$ — we get a unit. repr.



Uncertain reading. The loops in the underlined equation have a different shape. They are set here as ϕ.

Original source details: GELFAND-P000864

$$\varphi \longrightarrow V_\varphi.$$

Theorem 1. It is irreducible if $\dim X \geq 3$.

≥ 5 — Ismagilov
≥ 4 — Gelf., Graev, Vershik
≥ 3 — Albeverio and Høegh-Krohn.

Theorem 2. $\dim X = 1 \quad \Rightarrow$ the repr. is reducible

$$\dim X = 2 \qquad \text{— ?}$$

$b(t)$ — Brownian mot. in g. (standard; with resp. to the Killing form)
Consid. the eq.

$$\eta^{-1}(t)\, d\eta(t) = db(t), \qquad \eta \in C(\mathbb{R}^+, G). \tag{3}$$

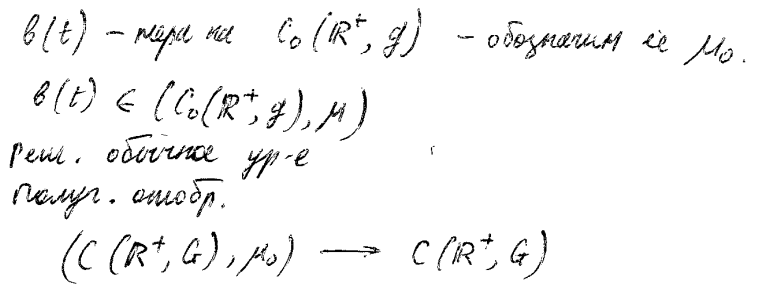
GELFAND-P000865

Explanation:
$b(t)$ — a measure on $C_0(\mathbb{R}^+, g)$ — denote it by μ_0.

$$b(t) \in \big(C_0(\mathbb{R}^+, g), \mu\big)$$

Solve the ordinary eq.
We obtain the map

$$\big(C(\mathbb{R}^+, G), \mu_0\big) \longrightarrow C(\mathbb{R}^+, G)$$

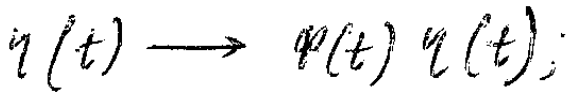


Original source details: GELFAND-P000865

Uncertain reading. The first formula has $C_0(\mathbb{R}^+, g)$ with μ. The domain of the map has $C(\mathbb{R}^+, G)$ with μ_0.

Stochastic — solve and transfer the measure.

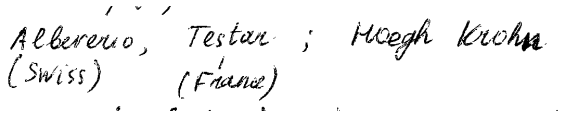
$$\eta(t) \longrightarrow \phi(t)\eta(t);$$



Uncertain reading. The closed-loop letter in these two formulas is provisionally set as ϕ.

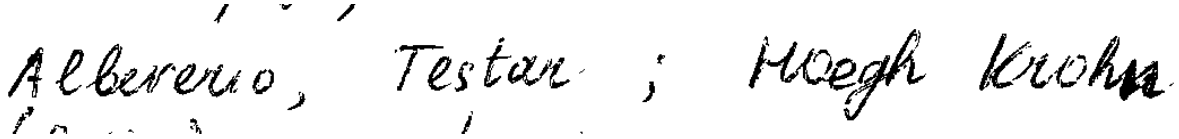
Original source details: GELFAND-P000865

$$\frac{d\mu(\phi\eta)}{d\mu(\eta)}$$



Original source details: GELFAND-P000865

Diagram and typed description.
Albeverio, Testar ; Hoegh Krohn.
(Swiss) (France)



Original source details: GELFAND-P000865

Uncertain reading. The first name is provisionally read as “Albeverio”. The second reads “Testar”. In the later 1981 article the names are printed as S. Albeverio, R. Høegh-Krohn and D. Testard.

$$L_2\big(C(\mathbb{R}^+, G), \mu\big), \qquad \varphi \in C^\infty(\mathbb{R}^+, G)$$



Uncertain reading. The open loop in the line with C^∞ is set as φ.

Original source details: GELFAND-P000865

$$\eta \longrightarrow \varphi\eta$$

$\varphi \longrightarrow W_\varphi$ we get a unit. repr. on $L_2(C(\mathbb{R}^+, G), \mu)$

Theorem $W_\varphi \cong V_\varphi$ equivalent.

The corresp. cocycle — the Maurer–Cartan form.

Factors:

$$[L_\infty \cup W_\varphi]'' \qquad \text{— a factor of type 3.}$$

GELFAND-P000866

22.10.79 Gelfand Sem.

Someone. (Baturin)



Uncertain reading. The handwritten name is provisionally read «Батурин» (“Baturin”). MCCME catalogue record 471 gives “I.A. Batalin”.

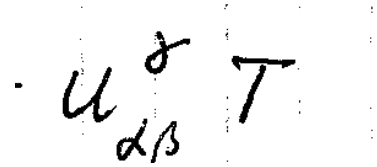
Original source details: GELFAND-P000866

On constraints.

First-class constraints

$$\begin{matrix} H \text{ — Hamiltonian}, & T_\alpha \text{ — constraints} \\ H, T_\alpha \text{ — funcs. of } q \text{ and } p. & \end{matrix} \tag{1}$$

$$\{T_\alpha, T_\beta\} = u^\gamma_{\alpha\beta} T, \qquad g^a = \begin{pmatrix} q^i \\ p_i \end{pmatrix}. \tag{2}$$



Original source details: GELFAND-P000866

Uncertain reading. No subscript is visible on T after $u^\gamma_{\alpha\beta}$.

$$\{A, B\} = \frac{\partial A}{\partial g^a} \varepsilon^{ab} \frac{\partial B}{\partial g^b}, \qquad \text{here } -\varepsilon^{ba} = \varepsilon^{ab} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{3}$$

First-class constraints — if condition (2) is satisfied, $u^\gamma_{\alpha\beta} = u^\gamma_{\alpha\beta}(p, q)$.

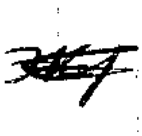
Question: can changes of var. be made so that $u^\gamma_{\alpha\beta} = \text{const}$. (4)

Rem. on the motivation: gauge transf. are gener. only by first-class constraints

Action:

$$S = \int dt \left[\frac{1}{2} g^a \varepsilon_{ab} \dot{g}^b - H - \lambda^\alpha T_\alpha \right]. \tag{5}$$

(λ^α — ordinary variables; [crossed out: ~~...~~] the extremum is now unconstrained)



Original source details: GELFAND-P000866

Undeciphered passage. The entry before "the extremum" is crossed out and illegible.

Gauge transf. are gener. by the constraints

[crossed out below: ~~$g^a \quad \delta g^a = \dots$~~]



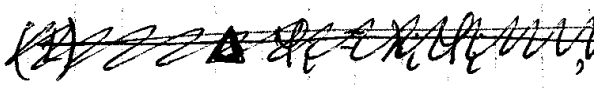
Original source details: GELFAND-P000866

Uncertain reading. The strokes under the hatching provisionally read g^a and $\delta g^a =$. The ending is illegible.

— • —

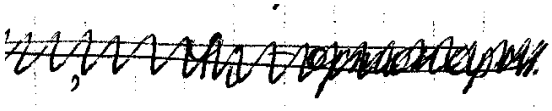
R. Kallosh. Quantum field theory

[line crossed out: ~~(1) $\Delta\phi_l = \lambda_l \phi_l, \dots$~~]



Original source details: GELFAND-P000866

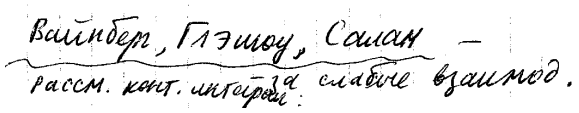
Uncertain reading. The left part of the crossed-out line provisionally reads (1) $\Delta\phi_l = \lambda_l \phi_l$. The subscripts are indistinct.



Original source details: GELFAND-P000866

Undeciphered passage. The right part after the comma is heavily hatched and illegible.

New Nobel laureates

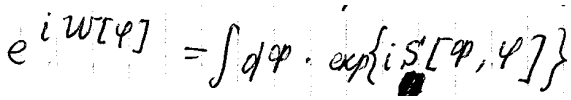


Original source details: GELFAND-P000866

Diagram and typed description.
Weinberg, Glashow, Salam —
for weak interact.

Symmetry underlies everything. Consid. the cont. integral.

$$e^{iW[\varphi]} = \int d\phi \cdot \exp\{iS[\phi, \varphi]\}. \tag{1}$$



Original source details: GELFAND-P000866

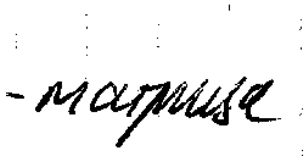
Uncertain reading. The open loop is set as φ (possibly ψ). The closed loop of the integration variable is set as ϕ.



Original source details: GELFAND-P000866

Undeciphered passage. The sign beneath S is blackened and illegible.

S is constructed from the class. action
from (1) much is determ.: probabilities, S-matrix.

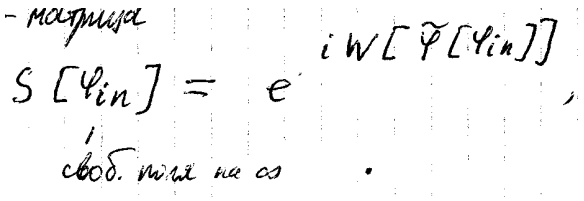


Original source details: GELFAND-P000866

Uncertain reading. The last letter in «матрице» (“matrix”) provisionally reads «е».

ϕ — quantum fluct. (one averages over them)
S-matrix

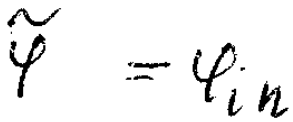
$$S[\varphi_{in}] = e^{iW[\tilde{\varphi}[\varphi_{in}]]}, \tag{2}$$



Original source details: GELFAND-P000866

Diagram and typed description.
φ_{in}
/
free fields at ∞.

$$\widetilde{\varphi} = \varphi_{in} + \int D^c(x - x') \frac{\delta \mathcal{L}_{int}}{\delta \widetilde{\varphi}(x')} \, d^4x'. \tag{3}$$



Uncertain reading. The looped sign in (3) and in the diagram is read as φ (possibly ψ).

Original source details: GELFAND-P000867

(Yang–Feldman eq.)

Iterative sol. of (3) gives trees.

D^c — Green's fn. for the free Klein–Gordon eq.

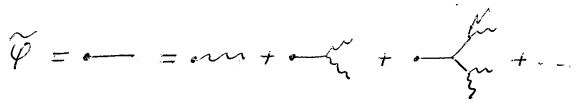


Diagram and typed description.

$\widetilde{\varphi} =$ —— $=$ ∿∿ $+$ —< $+$ >— $+ \cdots$

Original source details: GELFAND-P000867

Formula (2) contains amplitudes for prod. of any number of particles.

For gravity



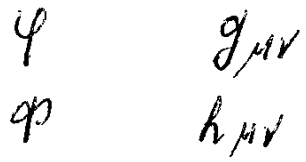
Uncertain reading. A sign, possibly $\mathcal{L}$, is cancelled before $S_{\text{кл}}$.

Original source details: GELFAND-P000867

$$S_{\text{кл}} = \int \mathcal{L} \, d^4x, \qquad \mathcal{L} = \sqrt{g} \, R[g_{\mu\nu}]. \tag{4}$$

S in (1) differs by quantum corrections

$$\begin{array}{ll} \varphi & g_{\mu\nu} \\ \phi & h_{\mu\nu} \end{array} \tag{5}$$



Uncertain reading. The upper sign is read as φ (possibly ψ), and the lower is rendered as ϕ.

Original source details: GELFAND-P000867

For gravity.

$$e^{iW[g_{\mu\nu}]} = \int dh \cdot e^{i\{S_{\text{кл}}[g_{\mu\nu}+h_{\mu\nu}]+S_{\text{кал}}+S_{\text{ф.ч.}}\}}\, d\bar{c}\, dc. \tag{6}$$

ф. ч. — fictitious particles.

$S_{\text{кл}}$ is given by formula (4) and is a functional of the metric g.

$h_{\mu\nu}$ — metric fluctuations.

$$S_{\text{кал.}} = \int \frac{1}{2}\left(D_\mu[g]h^\mu_\nu - \frac{1}{2}D_\nu[g]h^\mu_\mu\right)^2 \sqrt{g}\, d^4x. \tag{7}$$



Original source details: GELFAND-P000867

Uncertain reading. The argument of the second D runs into the beginning of h and is read as $[g]$.

D_μ — covariant deriv. for metric g

Raise and lower indices using g.

$$S_{\text{ф.ч.}} = \int \sqrt{g}\, \bar{c}^\nu D_\mu[g] D^\mu[g+h] c_\nu\, d^4x. \tag{8}$$

$S_{\text{ф.ч.}}$ is constructed from $S_{\text{кал}}$ so that the answer does not depend on $S_{\text{кал}}$ (principle of quantum relativity).

Observed by Feynman: without $S_{\text{кал}}$ there is no [cancelled] unitarity (total prob. $= 1$)



Original source details: GELFAND-P000867

Uncertain reading. The short cancellation before “unitarity” is illegible.

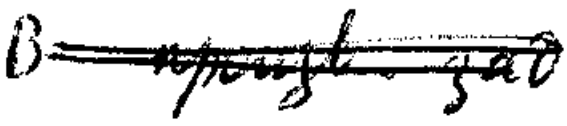


Original source details: GELFAND-P000867

Manuscript excerpt. The manuscript remark reads $S_{\text{кал}}$.

He [did so] in 1st approx.

De Witt, Popov, Faddeev — further.



Original source details: GELFAND-P000867

Uncertain reading. Cancelled Russian text: “В … зад”. The middle abbreviation is illegible, possibly “прибл.” (approx.).

In supergrav. one further term suffices.

GELFAND-P000868

[source number: 26]

In an arbitrary theory — infinitely many terms.

$$\underset{\text{(Einst.)}}{S_{\text{кл}}[g]} \qquad \text{inv. under transformations} \tag{9}$$

$$\delta g_{\mu\nu} = D_\mu \xi_\nu + D_\nu \xi_\mu, \qquad \xi_\mu \quad \begin{array}{l}\text{— arbitrary vector}\\ \text{(local fn. of } x\text{).}\end{array} \tag{10}$$

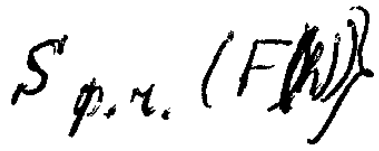
The quantum problem has an invariance:

$$W[g_{\mu\nu}] = W[g_{\mu\nu} + \delta g_{\mu\nu}], \qquad \delta g_{\mu\nu} \text{ taken from (10)} \tag{11}$$

In fact (6) needs further specification.

Another invariance law. Consider:

$$e^{iW_{\text{рад}}} \underset{\text{def}}{=} \int dh \cdot \exp i\Big\{ S_{\text{кл}}[g+h] - S_{\text{кл}}[g] - \frac{\delta S[g]}{\delta g_{\mu\nu}} h_{\mu\nu} + \\ + S_{\text{кал}}\big(F(h)\big) + S_{\text{ф.ч.}}\big(F(h)\big)\Big\}. \tag{12}$$



Original source details: GELFAND-P000868

Uncertain reading. The stroke before h has been retraced. The last argument is read as $F(h)$, possibly Fh.

F — a certain function satisfying certain conditions.

Gauge transf. [cancelled] preserve things with phys. meaning.



Original source details: GELFAND-P000868

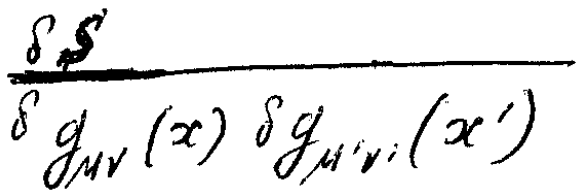
Uncertain reading. The cancelled word is illegible.

(The space remains isometric to the old one; same space, but I change the coord.).
Adding the terms $S_{\text{кал}}$, $S_{\text{ф.ч.}}$ is needed to handle degen. in the stationary-phase expansion.

$$W_{\text{рад}}[g_{\mu\nu}]_F = W_{F+\delta F}[g] - \frac{\delta S[g]}{\delta g_{\mu\nu}}\langle \delta h_{\mu\nu}\rangle \tag{13}$$

$S_{\text{кал}}$ is added for convergence (correcting the degeneracy)

$$\frac{\delta S}{\delta g_{\mu\nu}(x)\,\delta g_{\mu'\nu'}(x')}$$

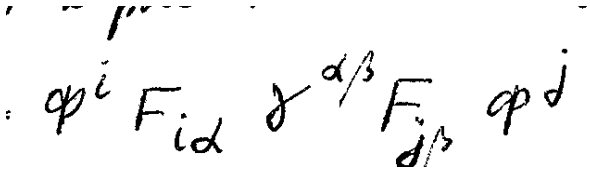


Uncertain reading. The numerator shows δS. No power on δ is discernible.

Original source details: GELFAND-P000868

Form of the terms that can be added:

$$S[\Phi] + \phi^i F_{i\alpha}\gamma^{\alpha\beta}F_{j\beta}\phi^j \tag{14}$$

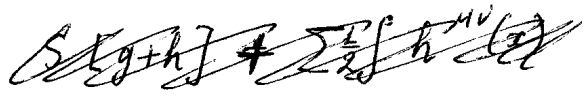


Uncertain reading. The two lowercase signs have two loops and are rendered as ϕ, as is the lower sign in (5).

Original source details: GELFAND-P000868

Let the external metric be flat

$$S[\Phi] + \frac{1}{2}\Phi^i F_{ij}\Phi^j \tag{15}$$



Uncertain reading. Cancelled line, read as $S[g+h] + \sum \frac{1}{2}\int h^{\mu\nu}(x)$. A cancellation stroke crosses the sign that resembles $\sum$.

Original source details: GELFAND-P000868

$$S[g+h] + \frac{1}{2}\int h^{\mu\nu}(x) F_{\mu\nu\mu'\nu'}(x,x',g) h^{\mu'\nu'}(x')\, d^4x\, d^4x' \tag{16}$$

F — a sufficiently arbitrary fn. of the points and the metric, for example:

$$\frac{1}{2}\left(D_\mu h^\mu_\nu - \frac{\beta}{2} D_\nu h\right)^2 \qquad \text{or} \qquad h^\mu h^\nu_\mu \tag{17}$$



Uncertain reading. After “or” the manuscript gives $h^\mu h^\nu_\mu$. No lower index on the first h is visible.

Original source details: GELFAND-P000869

Principle of relativity — it is independent of F. The result is a quadratic form

$$\frac{1}{2}\frac{\delta^2 S[g+h]}{\delta h_{\mu\nu}(x)\, \delta h_{\mu'\nu'}(x')} + F_{\mu\nu\mu'\nu'}(x,x') \tag{18}$$

The quadratic form must be nondegenerate. This is a condition on F.

The addition in (16) is the gauge-fixing action.

Read: Faddeev, Popov — review in UFN.



Uncertain reading. Two cancelled entries. The first may read “Теперь” (now). The lower one is illegible.

Original source details: GELFAND-P000869

In (13):

$\langle \delta h_{\mu\nu} \rangle$ — the change in the integral brought about by the given change of gauge.

$$\langle \delta h_{\mu\nu} \rangle = \int dh \cdot \exp i\{\widetilde{S}\}$$



Uncertain reading. The right-hand side ends with $\exp i\{\widetilde{S}\}$.

Original source details: GELFAND-P000869

$$\delta h_{\mu\nu} = D_\mu \xi_\nu + D_\nu \xi_\mu, \qquad \xi_\mu[\delta F] \tag{19}$$

Choose F such that

$$\frac{1}{2} h_{\mu\nu} F_{\mu\nu\mu'\nu'} h_{\mu'\nu'} = (h + \delta h)(F + \delta F) \cdot (h + \delta h). \tag{20}$$



Original source details: GELFAND-P000869

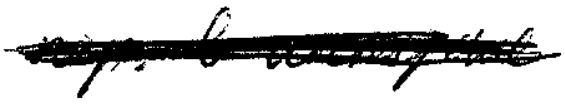
Uncertain reading. The factor $\frac{1}{2}$ is visible on the left-hand side only.

By construction:

$$W[g]_F = W[g + \delta g] \tag{21}$$

Another condition besides that. Correct construction of $S_{\text{ф.ч.}}$.

(13) is proved using a change of variables [cancelled] and precisely because of the correct choice of the action $S_{\text{ф.ч.}}$.



Original source details: GELFAND-P000869

Uncertain reading. The cancellation after "change" may read "пер. в интеграле" (variables in the integral).

In quantum electrodynamics.

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^2 \tag{22}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{23}$$

$$\delta A_\mu = \partial_\mu \xi, \qquad \xi = \xi(x) \tag{24}$$



Original source details: GELFAND-P000869

Uncertain reading. The ξ after ∂_μ has been retraced. The underlying strokes are illegible.



Original source details: GELFAND-P000869

Uncertain reading. An equality is cancelled between the parts of (24), possibly $\Lambda = \Lambda(x)$ or $\lambda = \lambda(x)$.

Nothing changes under this — $F_{\mu\nu}$ and $\mathcal{L}$ are unchanged.

$$\int dA_\mu \cdot \exp i\{S_{\text{кл}}[A] + S_{\text{кал}}[A] + S_{\text{ф.ч.}}\} \tag{25}$$

GELFAND-P000870

[source number: 27]

Choice of gauge-fixing term

$$S_{\text{кал}} = \frac{1}{2\alpha}(\partial_\mu A_\mu)^2 \tag{26}$$

or

$$\frac{1}{2\alpha}(\partial_\mu A_\mu + \beta n_\mu A_\mu)^2 \tag{27}$$



Uncertain reading. The second digit has been heavily retraced and is read as 7. The underlying strokes are illegible.

Original source details: GELFAND-P000870

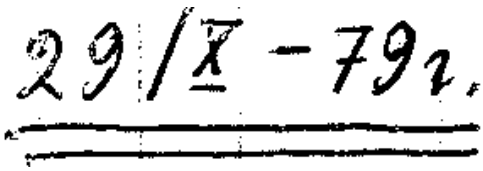
Quadr. form in the classical action:

$$\frac{1}{2}A_\mu(\Box g_{\mu\nu} - \partial_\mu\partial_\nu)A_\nu \tag{28}$$

GELFAND-P000871

[source number: 31]

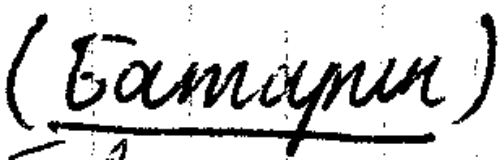
29/X-79 Gelfand Sem.



Uncertain reading. The manuscript gives 29/X-79. The MCCME catalogue dates record 473 “29-Sep-79”.

Original source details: GELFAND-P000871

Physicist (the same one). (Baturin)



Uncertain reading. The manuscript appears to read “Батурин” (Baturin). The MCCME catalogue gives I.A. Batalin.

Original source details: GELFAND-P000871

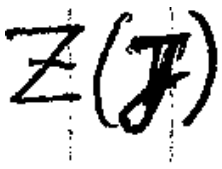
On quantizing gauge theories by means of functional integrals.

Consider the integral

$$Z(J) = c\int e^{iW(\varphi)+i\int J(x)\varphi(x)\,d^4x}\,d\mu(\varphi) \tag{1}$$

(J — “source”).

(Fourier transf. of a functional.)



Uncertain reading. The retraced argument of Z is read as J.

Original source details: GELFAND-P000871



Original source details: GELFAND-P000871

Uncertain reading. The open-loop sign is rendered as φ (possibly ψ).



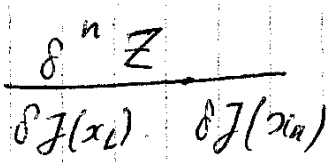
Original source details: GELFAND-P000871

Uncertain reading. The retraced argument of μ is read as φ. The underlying strokes are illegible.

(derivative of the functional)

$$\frac{\delta^n Z}{\delta J(x_1) \quad \delta J(x_n)} = G(x_1, \ldots, x_n) \quad \text{— Green's fn.} \tag{2}$$

(everything is expressed through it.)

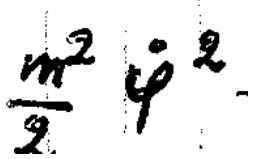


Original source details: GELFAND-P000871

Uncertain reading. The denominator shows $\delta J(x_1)$ and $\delta J(x_n)$, with a space between them.

$$W(\varphi) = \text{classical action of the field} = \int \mathcal{L}\, d^4x \tag{3}$$

$$\mathcal{L} = -\frac{1}{2}(\partial_\mu \varphi)^2 - \frac{m^2}{2}\dot{\varphi}^2 - \lambda\varphi^4 \tag{4}$$



Original source details: GELFAND-P000871

Uncertain reading. A separate dot is visible above the letter in the second term. It is set as $\dot{\varphi}^2$. The meaning of the dot is unclear.

(for example)

(1) — over all fields φ with certain decay at ∞.

Interpretation of (1): a) expansion in λ into a perturbation series

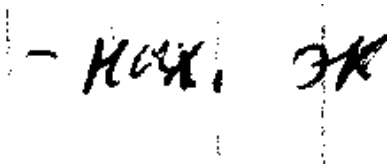
(in grav. — small departures from flat space)

b) evaluation by stationary phase near a classical extremal.

a) is currently inapplicable to grav..

b) better — accounts for more complicated classical fields at once.

b) — certain extremals of the full action.



Original source details: GELFAND-P000871

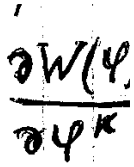
Uncertain reading. The abbreviation before "extremals" is provisionally read as "нек." (certain).

Stationary phase gives the asymptotics of

$$\int e^{\frac{i}{\hbar}W(\varphi)}(d^n\varphi) \tag{5}$$

as $\hbar \to 0$

$$\frac{\partial W(\varphi)}{\partial \varphi^\kappa} = 0 \quad \text{— stationary points.} \tag{6}$$



Uncertain reading. The superscript is rendered as κ, possibly Latin k.

Original source details: GELFAND-P000871

At the stationary point $\bar{\varphi}$, expand by Taylor's formula

$$\text{Leading term:} \qquad e^{\frac{i}{\hbar}W(\bar{\varphi})}\frac{1}{\sqrt{\operatorname{Det}\left\{\dfrac{\partial^2 W(\bar{\varphi})}{\partial\varphi\,\partial\varphi}\right\}}}(1+o(\hbar)) \tag{7}$$

Then everything reduces to a Gaussian integral, answer $\dfrac{1}{\sqrt{\det A}}$.

A — degenerate $\Rightarrow$ fails. This is what arises in gauge theories.

GELFAND-P000872

Example:

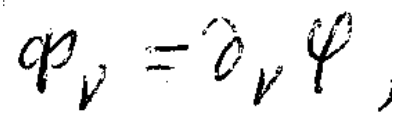
in electrodynamics

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{8}$$

Quadr. form:

$$\frac{1}{2}A_\mu \underbrace{(\Box\delta_{\mu\nu} - \partial_\mu\partial_\nu)}_{K_{\mu\nu}} A_\nu \tag{9}$$

$$\text{if} \quad \phi_\nu = \partial_\nu\varphi, \quad \text{then} \quad K_{\mu\nu}\phi_\nu = 0 \quad \text{and cannot be inverted.} \tag{10}$$

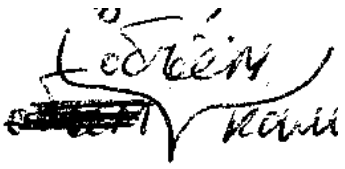


Uncertain reading. The two letter shapes in (10) are distinguished. The form with two loops is set as ϕ. The form with an open left loop is set as φ (possibly ψ).

Original source details: GELFAND-P000872

The same in gravity, Y.–M., etc.

We have to factor out [cancelled] [above the line: the volume] of the gauge group.

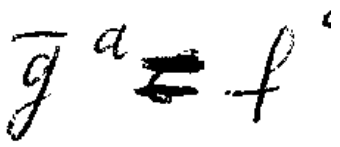


Uncertain reading. The cancelled word before the inserted "volume" is illegible. It may be the same word.

Original source details: GELFAND-P000872

$g^a \in M$ — space
and let a group act

$$g^a \longrightarrow \bar{g}^a = f^a(g,\theta), \qquad \theta \text{ — group parameters.} \tag{11}$$



Uncertain reading. The heavily retraced sign between $\bar{g}^a$ and f^a is read as equality.

Original source details: GELFAND-P000872

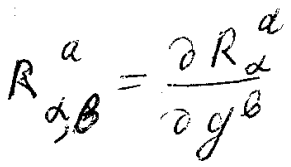
$$\int e^{iW(g)}[dg] \tag{12}$$

$W(g)$ is invariant
Consider a transf. (infinitesimal)

$$\delta g^a = R^a_\alpha(g)\theta^\alpha, \qquad \theta^\alpha \to 0 \tag{13}$$

$$R^a_\alpha(g) = \left.\frac{\partial f^a(g,\theta)}{\partial \theta^\alpha}\right|_{\theta=0} \tag{14}$$

$$R^a_{\alpha,b} = \frac{\partial R^a_\alpha}{\partial g^b} \tag{15}$$



Uncertain reading. The final lower index on the left has been retraced and is read as b, following the index in the denominator.

Original source details: GELFAND-P000872

Condition:

$$R^a_{\alpha,a} = 0 \qquad \text{(unimodularity} \Rightarrow \text{left and right measures coincide)} \tag{16}$$

Lie eqs.

$$\frac{\partial \bar{g}^a}{\partial \theta^\beta} = R^a_\alpha(\bar{g}) \underbrace{\lambda^\alpha_\beta(\theta)}_{\text{group frame}}, \tag{17}$$

there are also $t^\alpha_{\beta\gamma}$ — structure consts. of the Lie alg.
Group composition law

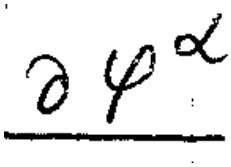
$$f^a(f(g,\theta),\theta') = f^a(g,\varphi(\theta,\theta')) \tag{18}$$

GELFAND-P000873

[source number: 32]

$$\lambda = \mu^{-1}, \quad \text{where} \tag{19}$$

$$\mu^{\alpha}_{\beta}(\theta) = \left.\frac{\partial \varphi^{\alpha}}{\partial \theta'^{\beta}}\right|_{\theta'=0}. \tag{19}'$$



Uncertain reading. The sign in the numerator is rendered as φ, as in (18). The manuscript uses a distinct shape for ψ in (20) and the following equations.

Original source details: GELFAND-P000873

Consider a hypersurface in M.

$$\psi^{\alpha}(f(g,\theta)) = 0 \tag{20}$$

We want $\forall g\ \exists!$ sol. θ of eq. (20). (cross-section!)

The class of admissible gauges is where this condition holds. We have:

$$1 = \int [d\theta]\, \delta(\psi(f(g,\theta))\, \mathrm{Det}\left\{\frac{\partial \psi(f(g,\theta))}{\partial \theta}\right\} \tag{21}$$



Uncertain reading. Two closing parentheses follow the argument of f, with three opening parentheses in the expression.

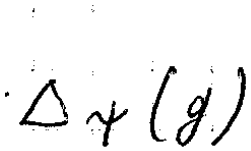
Original source details: GELFAND-P000873

Compute the Jacobian from the Lie eqs.:

$$\frac{\partial \psi^{\alpha}(f(g,\theta))}{\partial \theta^{\gamma}} = \psi^{\alpha}_{,a}(f(g,\theta)) R^{a}_{\beta}(f(g,\theta)) \lambda^{\beta}_{\gamma}(\theta) \tag{22}$$

Hence

$$\mathrm{Det}\left\{\frac{\partial \psi}{\partial \theta}\right\} = \Delta_{\psi}(g)\, \mathrm{Det}\{\lambda(\theta)\} \tag{23}$$



Uncertain reading. Equation (23) gives $\Delta_{\psi}(g)$ with argument g.

Original source details: GELFAND-P000873

Here

$$\Delta_\psi(g) = \mathrm{Det}\,\mathcal{D}^{-1} \tag{24}$$

$$\mathcal{D}^{-1}{}^{\alpha}_{\beta}(g) = \psi^{\alpha}_{,a}(g) R^{a}_{\beta}(g) \tag{25}$$

Substitute the Jacobian in formula (21). We obtain:

$$1 = \int \delta(\psi(f(g,\theta))\Delta_\psi(f(g,\theta)) \underbrace{d\sigma(\theta)}_{\substack{\text{invariant measure} \\ \text{on the group}}} \tag{26}$$



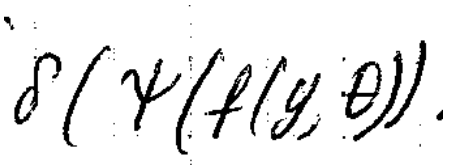
Original source details: GELFAND-P000873

Uncertain reading. In (26), two closing parentheses follow the argument of f, as in (21).

$$d\sigma(\theta) = \mathrm{Det}\{\lambda(\theta)\}[d\theta] \quad \text{— invariant measure on the group.} \tag{27}$$

Return to the integral (12)

$$\begin{aligned} &\int e^{iW(g)}[dg] \\ &\qquad = \int e^{iW(g)} \int \delta(\psi(f(g,\theta))\Delta_\psi(f(g,\theta))\, d\sigma(\theta)[dg] = \\ &\qquad = \int d\sigma(\theta) \int e^{iW(g)} \delta(\psi(f(g,\theta))\Delta_\psi(f(g,\theta))[dg] \end{aligned} \tag{28}$$



Original source details: GELFAND-P000873

Uncertain reading. The upper line of (28) shows two closing parentheses after the argument of f in the delta expression.



Original source details: GELFAND-P000873

Uncertain reading. The lower line of (28) uses the same notation.

Change of variable;

$$\underset{\text{old var.}}{g^a} \longrightarrow \underset{\text{new var.}}{f^{-1\,a}(g,\theta)} \tag{29}$$

θ is fixed here (the same one over which we integrate)

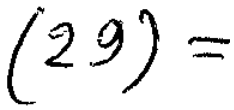
$W(g)$ inv.; $[d\sigma]$ inv. $\Rightarrow$ they will stay the same.



Uncertain reading. The manuscript gives $[d\sigma]$ as invariant after $W(g)$.

Original source details: GELFAND-P000874

$$\psi(f(g,\theta)) \rightsquigarrow \psi(g)$$
$$\Delta_\psi(f(g,\theta)) \rightsquigarrow \Delta_\psi(g)$$
$$(29) = \int d\sigma(\theta) \int e^{iW(g)} \delta(\psi(g)) \Delta_\psi(g)[dg] \qquad (30):$$



Uncertain reading. The left-hand side of (30) reads (29).

Original source details: GELFAND-P000874

Now we divide by $\int d\sigma(\theta)$ (drop it)

On the hypersurf. [cancelled] $\psi(g) = 0$ the quadr. form is already nonsingular.



Uncertain reading. The cancellation is illegible. It may be $\psi(g)$.

Original source details: GELFAND-P000874

the cross-section was

$$\psi^\alpha(g) = 0 \tag{31}$$

Now replace

$$\psi^\alpha(g) \longrightarrow \psi^\alpha(g) - c^\alpha \tag{32}$$

and obtain a cross-section.

$$\psi^\alpha(g) = c^\alpha. \tag{33}$$

This gives

$$\int e^{iW(g)} \delta(\psi(g) - c) \Delta_\psi(g)[dg] \tag{34}$$

and consider

$$\int \Big(\int e^{iW(g)} \delta(\psi(g) - c) \Delta_\psi(g)[dg]\ e^{i\Omega(c)}[dc] \Big) \tag{35}$$

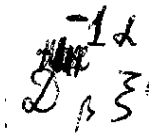
By construction it is independent of ψ and therefore of $c \Rightarrow$ integration changes nothing $\Omega(c)$ — any fn. of c. Hence

$$(35) = \int e^{iW(g) + i\Omega(\psi(g))} \Delta_\psi(g)[dg] \tag{36}$$

(we have integrated over c)

We have:

$$\Delta_\psi(g) = \mathrm{Det}\, \mathcal{D}^{-1}(g) = \int [d\bar{\xi}][d\xi] e^{i\bar{\xi}_\alpha \mathcal{D}^{-1}{}^\alpha_\beta \xi^\beta} = \tag{37}$$



Original source details: GELFAND-P000874

Uncertain reading. Above $\mathcal{D}$, superscripts are cancelled and $-1\,\alpha$ is inserted. The cancelled marks are illegible.



Original source details: GELFAND-P000874

Uncertain reading. After the last equality, $\int [d\bar{\xi}]$ and a second factor are cancelled. The second factor may be $[d\xi]$ or $[dg]$.

GELFAND-P000875

[source number: 33]

$$= \int [d\bar{\xi}][d\xi] e^{i\bar{\xi}_\alpha \psi^\alpha_{,a} R^a_\beta \xi^\beta}$$

ξ, $\bar{\xi}$ — anticommuting Grassmann variables

Now substitute in (36):

$$(36) = \int e^{iW_{\text{eff}}(g,\bar{\xi},\xi)} [dg][d\bar{\xi}][d\xi], \tag{38}$$

$$W_{\text{eff}} = W(g) + \Omega(\psi(g)) + \bar{\xi}_\alpha \psi^\alpha_{,a}(g) R^a_\beta(g) \xi^\beta \tag{39}$$

It is in fact independent of ψ by construction.

It does depend on Ω. But

$$\frac{(36)}{\int e^{i\Omega(c)} dc} \qquad \text{is independent of } \Omega. \tag{40}$$

W_{eff} is, of course, now noninvariant.

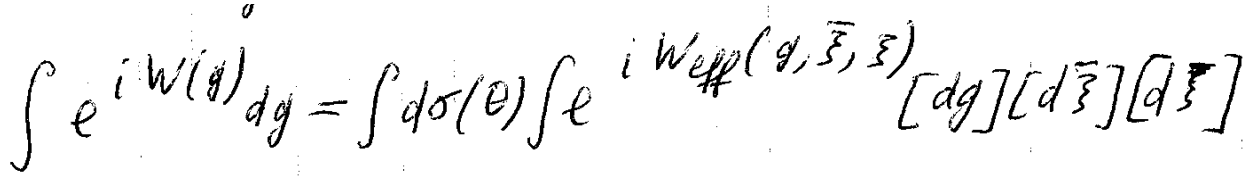
But it exhibits deeper invariance properties

(BRS — invariance)

or BRS — symmetry

Thus: it has been proved that

$$\int e^{iW(g)} dg = \int d\sigma(\theta) \int e^{iW_{\text{eff}}(g,\bar{\xi},\xi)} [dg][d\bar{\xi}][d\xi] \tag{41}$$



Original source details: GELFAND-P000875

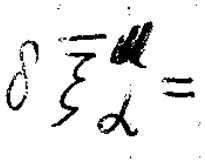
Uncertain reading. The right-hand side of (41) has no denominator from (40).

Reminder. We had

$$\delta g^a = R^a_\alpha(g) \theta^\alpha, \quad \theta^\alpha \to 0, \tag{42}$$

Forget that. Instead:

$$\begin{cases} \delta g^a = R^a_\alpha(g) \xi^\alpha \mu \\ \delta \xi^\alpha = \frac{1}{2} t^\alpha_{\beta\gamma} \xi^\beta \xi^\gamma \mu \\ \delta \bar{\xi}_\alpha = \Omega_{,\alpha}(\psi(g)) \mu \end{cases} \tag{43}$$



Uncertain reading. The superscript on $\bar{\xi}$ is cancelled, possibly α or a. The subscript is α.

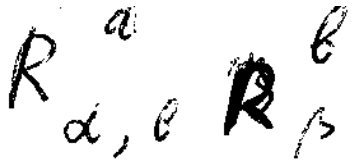
Original source details: GELFAND-P000875

μ — anticommuting Grassmann param.;

$$\mu^2 = 0, \quad \mu\xi = -\xi\mu, \quad \mu\bar{\xi} = -\bar{\xi}\mu, \quad \xi\bar{\xi} = -\bar{\xi}\xi. \tag{44}$$

To check invariance, use the ordinary Lie relations for R:

$$R^a_{\alpha,b} R^b_\beta - (\alpha \leftrightarrow \beta) = -t^\gamma_{\alpha\beta} R^a_\gamma \tag{45}$$



Uncertain reading. The last lower index on the first R resembles e. It is read as b, matching the superscript of the second R.

Original source details: GELFAND-P000875

GELFAND-P000876

We can now work directly with expr. (38), bearing in mind that it has BRS symmetry.
In modern theories, conversely, W_{eff} is sought from considerations of BRS symmetry.
ξ-fields — ghosts; they have opposite statistics (Bose $\leftrightarrow$ Fermi and vice versa).
Independence of ψ and Ω — quantum gauge invariance.



Undeciphered passage. The small cancelled sign above (46) is illegible.

Original source details: GELFAND-P000876

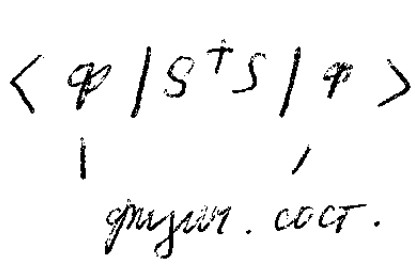
$$W_{\text{eff}} \rightsquigarrow W_{\text{eff}} + J_a g^a \tag{46}$$

gives the generating functional. ~~This~~. S-matrix only in the g sector
S-matrix turns out to be unitary
Ghosts never appear externally — there are no sources.
Add sources for the ghosts.
S^+S unit. on the full space.
In $\langle \phi \mid S^+S \mid \phi \rangle$ the contributions from nonphysical degrees of freedom of gauge [cancelled] fields and ghosts cancel.



Uncertain reading. The entry after “кал.” is cancelled. It may read “по”.

Original source details: GELFAND-P000876



$\langle\ \phi|\ S^+S\ |\phi\ \rangle$
phys. states

Diagram and typed description.

Original source details: GELFAND-P000876

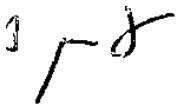
Rem. on the Lie alg.

$$\Gamma_\alpha = R^a_\alpha \frac{\partial}{\partial g^a}, \tag{47}$$

$$[\Gamma_\alpha, \Gamma_\beta] = t^\gamma_{\alpha\beta}\Gamma_\gamma \tag{48}$$

hence

$$2(\Gamma\xi)(\Gamma\xi) = t^\gamma_{\alpha\beta}\xi^\alpha\xi^\beta\Gamma^\gamma \tag{49}$$



Uncertain reading. At the end of (49), γ is written as a superscript on Γ.

Original source details: GELFAND-P000876

$[dg][d\bar{\xi}][d\xi]$ — is invariant under the BRS transf..

How do we check independence of ψ?

Choose μ in a special way:

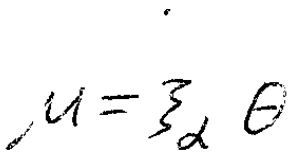
$$\mu = \bar{\xi}_\alpha \delta\psi^\alpha \tag{50}$$

and perform a BRS transformation.

The integral's independence of coordinates gives independence of ψ.

For

$$\mu = \xi_\alpha\theta \tag{51}$$



Uncertain reading. In (51), ξ has no bar and θ has no index.

Original source details: GELFAND-P000876

we obtain Ward identities, the eqs. of renormalization theory.

Bilinearity in the ghosts follows because the gauge transf. form a group.

GELFAND-P000877

[source number: 34]

In supergravity, gauge transf. do not form a group.

Yang–Mills.

Let a semisimple compact Lie group be given

t^{abc} — struct. constants of the Lie alg.

and

$$t^{abc} = (t^b)^{ac}, \quad t^b \text{ — matrix}, \quad t^b \text{ — generator of the adjoint rep.} \tag{52}$$

$$[t^a, t^b] = t^{abc} t^c \qquad \text{(assume an orthogonal basis)} \tag{53}$$

$$A_\mu(x) = A^a_\mu(x) t^a \tag{54}$$

g^a — includes μ, A and x.

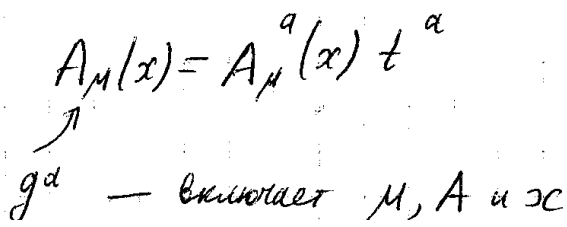


Diagram and typed description. $A_\mu(x)$ over g^a

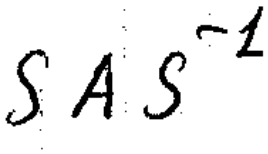
Original source details: GELFAND-P000877

Y.–M. Lagr.

$$\mathcal{L}_{\text{Y.M.}} = -\frac{1}{4} G^a_{\mu\nu} G^a_{\mu\nu} \tag{55}$$

$$G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g t^{abc} A^b_\mu A^c_\nu \tag{56}$$

$$\bar{A}_\mu = S A S^{-1} - \frac{1}{g} (\partial_\mu S) S^{-1} \tag{57}$$



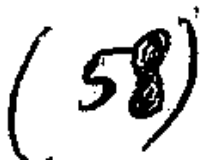
Uncertain reading. The first term of (57) reads SAS^{-1}, with no index on A.

Original source details: GELFAND-P000877

$S(x) = S$ — matrix from the adjoint rep. of the group

infinitesimal gauge transf.

$$\delta A^a_\mu(x) = \nabla^{ab}_\mu(A) \theta^b(x) \tag{58}$$

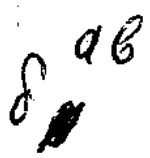


Uncertain reading. The 8 in equation number (58) is retraced.

Original source details: GELFAND-P000877

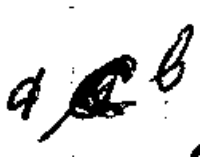
Here

$$\nabla^{ab}_{\mu}(A) = \partial_{\mu}\delta^{ab} + gt^{acb}A^{c}_{\mu} \quad \text{— covariant derivative.} \tag{59}$$



Uncertain reading. There is an illegible cancelled mark below δ^{ab}. It may be μ.

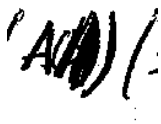
Original source details: GELFAND-P000877



Uncertain reading. The middle superscript on t is overwritten. It may be c as in t^{acb}. The underlying mark is illegible.

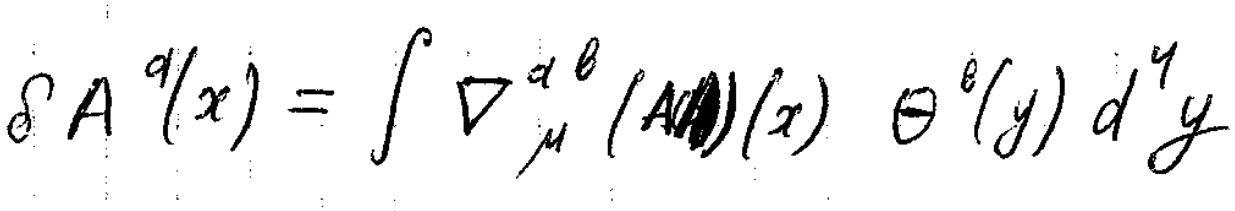
Original source details: GELFAND-P000877

$$\delta A^{a}(x) = \int \underset{\text{analogue of } R}{\nabla^{ab}_{\mu}(A)(x)}\ \theta^{b}(y)d^{4}y \tag{60}$$



Uncertain reading. There is a cancelled entry in parentheses after A. It may be (x) or (y).

Original source details: GELFAND-P000877



Original source details: GELFAND-P000877

Uncertain reading. On the left of (60), A^{a} has no subscript μ. No separate factor connecting x and y is written.

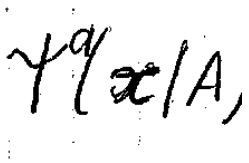
correspondence with the old notation

$$\begin{cases} \theta \text{ — parameter} \\ a \text{ — is } \mu, A, x. \\ \alpha \text{ — is } y \text{ and } b. \end{cases}$$

Now write out:

W_{eff}

$$\psi^{a}(x \mid A) = \partial_{\mu}A^{a}_{\mu}(x) \tag{61}$$



Uncertain reading. The x in the left-hand argument of (61) is retraced.

Original source details: GELFAND-P000877

$$\Omega(\psi) = -\frac{\alpha}{2}\int (\psi^{a})^{2}d^{4}x \tag{62}$$

Eff. Lagrangian

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{Y.M.}} - \frac{\alpha}{2}(\partial_\mu A^a_\mu)^2 - (\partial_\mu \bar{\xi}^a)\nabla^{ab}_\mu(A)\xi^b. \tag{63}$$

Next time 12 November.

GELFAND-P000878

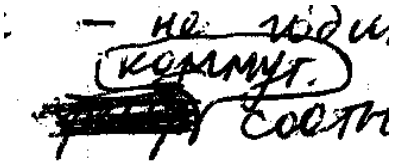
12/XI-79. Gelfand Sem..

I. M. G. Following Hawking

There are 3 approaches to quantizing GR

1) operators — unsuitable, since one must lower and raise ind.; noncomm.

2) canonical commut. rel. — poor, since space has to be foliated — against the spirit of GR



Original source details: GELFAND-P000878

Uncertain reading. "commut." is inserted above an illegible cancelled word and circled.

— and take $M^4 = M^3 \times R$, whereas all topologies are needed.



Original source details: GELFAND-P000878

Uncertain reading. There is a cancelled sign before the last line. It may read "3)".



Original source details: GELFAND-P000878

Uncertain reading. There is a cancelled word before "Points". The first letter may be "K".

Points $\to$ uncertainty princ.

GELFAND-P000879

[source number: 38]

Consider

$$\frac{1}{16\pi G}\int (R - 2\Lambda)(-g)^{1/2}\, d^4x + \int L_m(-g)^{1/2}\, d^4x \tag{1}$$

Λ — cosmological term, $\Lambda = \text{const}$.

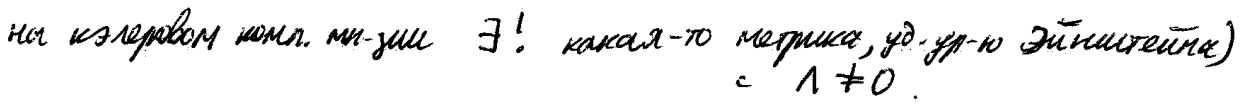
([possibly "naïv.:"] Yau's thm.: on a compact Kähler manif. $\exists!$ some metric satisfying Einstein's eq.)

[Added below the line: with $\Lambda \neq 0$.]



Original source details: GELFAND-P000879

Uncertain reading. The word before "Yau's thm." is unclear. It may read "наивн:" ("naïv.:").



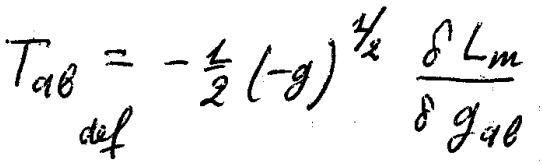
Original source details: GELFAND-P000879

Uncertain reading. The manuscript gives ∃! and $\Lambda \neq 0$. The second condition is added below the line.

L_m — matter-field Lagrangian
(functional of the metric and extra fields)

$$R_{ab} - \frac{1}{2} g_{ab} R + \Lambda g_{ab} = 8\pi G T_{ab}, \qquad \text{— Einstein's eq.} \tag{2}$$

$$T_{ab} \underset{\text{def}}{=} -\frac{1}{2}(-g)^{1/2} \frac{\delta L_m}{\delta g_{ab}} \qquad \text{— stress–energy tensor} \tag{3}$$



Uncertain reading. Equation (3) has coefficient $-1/2$, exponent $1/2$ and lower indices on g_{ab} in the manuscript.

Original source details: GELFAND-P000879

(the most correct method,)
devised by Hilbert
(4) Boundary conditions: something $= 0$ on the boundary.
Gibbons, Hawking (1977) added a boundary term

$$\frac{1}{8\pi G} \int K (\pm h)^{1/2} \, d^3x + C \tag{5}$$

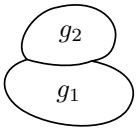
K — trace of the 2nd quadr. form of the surface,
C depends only on the boundary metric h (induced)
and is independent of g at interior points
(see also Landau and Lifshitz)
We need:

$$I[g_1 + g_2] = I[g_1] + I[g_2] \tag{6}$$



Original source details: GELFAND-P000879

Diagram and typed description.

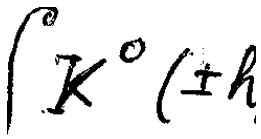


This will hold if we add term (5).

awkward — clumsy (неуклюжий)

Then

$$C = -\frac{1}{8\pi G}\int K^0(\pm h)^{1/2}\,d^3x \tag{7}$$



Original source details: GELFAND-P000879

Uncertain reading. A small oval is written above K in (7). It may be 0 or $\circ$.



Original source details: GELFAND-P000879

Uncertain reading. There is a cancelled entry below (7). The initial sign may be Z. The full reading remains uncertain.

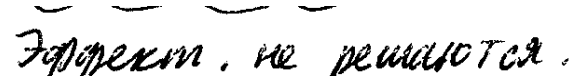
GELFAND-P000880

Batalin.

Quantization of systems with constraints.

2 classes of constraints: 1st and 2nd class.

Eff. they are not solved.



Original source details: GELFAND-P000880

Uncertain reading. The preferred reading is “решаются” (plural). The singular “решается” is also possible.

One must write a Hamiltonian path integral with constraints

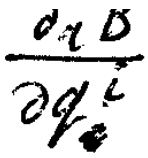
$$q^i, p_i \text{ — can. coordinates}, \qquad M \text{ — phase space} \tag{1}$$

$i = 1, \ldots, n$.

q^i, p_i can be even or odd. There is a bracket

$$H_0(q,p) \text{ — Hamiltonian}, \quad T_\alpha(q,p) \text{ — constraints}, \quad \alpha = 1, \ldots, 2m. \tag{2}$$

$$\{A,B\} = \frac{\partial_r A}{\partial q^i}\frac{\partial_\ell B}{\partial p_i} - (-1)^{n_A n_B}\frac{\partial_r B}{\partial q^i}\frac{\partial_\ell A}{\partial p_i} \tag{3}$$



Original source details: GELFAND-P000880

Uncertain reading. A subscript on the second q^i is cancelled. It may be i.

n_A, n_B — parities.

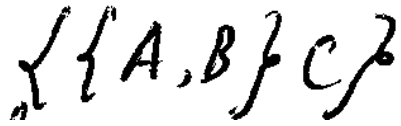
$$\{B, A\} = -(-1)^{n_A n_B}\{A, B\} \tag{4}$$

hence

$$\{A, A\} = -(-1)^{n_A}\{A, A\} \tag{5}$$

BRS — symmetry
(Becchi, Rouet, Stora).

$$(-1)^{n_A n_C}\{\{A, B\}C\} + \text{cycl. } (A, B, C) = 0 \tag{6}$$



Uncertain reading. In (6) no separate comma is visible between the inner brace and C.

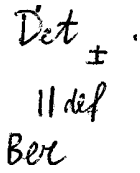
Original source details: GELFAND-P000880

Upon quantization:

$$\{A, B\} \mapsto [A, B] = AB - (-1)^{n_A n_B} BA \tag{7}$$

Definition

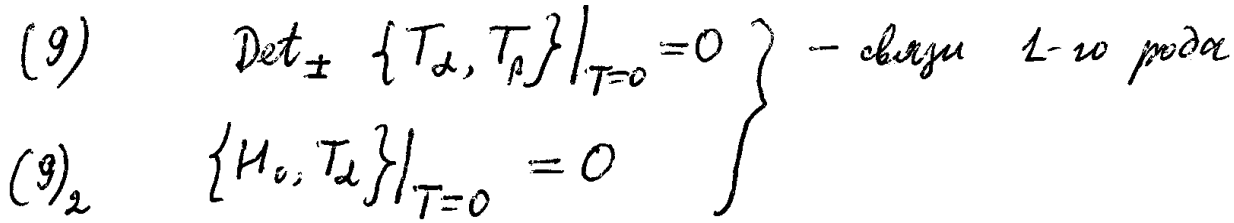
$$\underset{\substack{\| \text{ def} \\ \text{Ber}}}{\text{Det}_{\pm}\{T_\alpha, T_\beta\}}\Big|_{T=0} \neq 0 \quad \text{— 2nd-class constraints} \tag{8}$$



Uncertain reading. In (8), $\|$ def is added below $\text{Det}_{\pm}$, with Ber below it.

Original source details: GELFAND-P000880

$$\left.\begin{array}{ll} (9) & \text{Det}_{\pm}\{T_\alpha, T_\beta\}|_{T=0} = 0 \\ (9)_2 & \{H_0; T_\alpha\}|_{T=0} = 0 \end{array}\right\} \quad \text{— 1st-class constraints}$$



Original source details: GELFAND-P000880

Uncertain reading. The first line of (9) sets the determinant to zero. The second line is labelled $(9)_2$.

quant. of 1st-class constraints reduces to 2nd-class constraints (see below)

Consider 2nd-class constraints.

Class. dynamics — finding a constrained extremum of the class. action

$$S[q, p] = \int dt\,(p_i \dot{q}^i - H_0)\Big|_{T=0} \tag{10}$$

This is equiv. to an unconstrained extremum problem:

GELFAND-P000881

[source number: 39]

$$S[q,p,\lambda] = \int dt\,(p_i\dot{q}^i - H_0 - T_\mu\lambda^\mu) \tag{11}$$

Dirac bracket:

$$\{A,B\}_D \underset{\text{def}}{=} \{A,B\} - \{A,T_\alpha\}(Q^{-1})^{\alpha\beta}\{T_\beta,B\} \tag{12}$$

Here:

$$Q_{\alpha\beta} = \{T_\alpha, T_\beta\} \tag{13}$$

(I. M. G.: this is the bracket constr. from the restriction of the sympl. form to the submanif.). Consider the eq. for extremals of this action:

$$\begin{cases} \dot{q}^i = \{q^i, H_0\}_D \\ \dot{p}_i = \{p_i, H_0\}_D \\ \lambda^\alpha = -(Q^{-1})^{\alpha\beta}\{T_\beta, H_0\} \end{cases} \tag{14}$$



Original source details: GELFAND-P000881

Uncertain reading. A subscript is inked out after the final bracket in (14). It may be D.

One obtains the stand. Hamilton eqs. The Dirac bracket plays the role of the bracket. Hence it is precisely this bracket that is to be [with] replaced with the commutator.



Original source details: GELFAND-P000881

Uncertain reading. The manuscript gives an extra “на” (“with”) before “заменять” (“replace”).

From eq. [cancelled]

$$T_\alpha(q,p) = 0 \tag{15}$$



Original source details: GELFAND-P000881

Uncertain reading. A sign is cancelled between “Из” (“From”) and “ур-я” (“eq.”). It may be T.

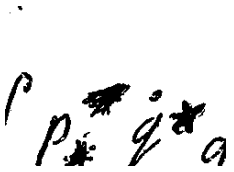
express

$$q^i = q^i(q^*, p^*), \quad p_i = p_i(q^*, p^*) \tag{16}$$

$* = 1, \ldots, n - m$

We want q^*, p_* to be canonical, i.e.

$$\int p_i \dot{q}^i \, dt \Big|_{T=0} = \int p_* \dot{q}^* \, dt \tag{17}$$



Original source details: GELFAND-P000881

Uncertain reading. On the right in (17), a superscript on p is cancelled, possibly an asterisk. The subscript asterisk is retained.

The eq. will take the form:

$$\dot{q}^* = \{q^*, H^{\text{phys}}\} \qquad \text{and likewise for } p. \tag{18}$$

$$H^{\text{phys}}(q^*, p_*) = H_0(q, p)|_{T=0} \tag{19}$$

This follows from the usual prop. of the Poisson bracket

$$\{A, B\}_D|_{T=0} = \text{[cancelled]} \quad \{A^*, B^*\}, \tag{20}$$



Original source details: GELFAND-P000881

Uncertain reading. After the equals sign a formula is cancelled, tentatively $\sum_* \frac{\partial A^*}{\partial q^*}$.

where

$$A^*(q^*, p^*) = A\big(q(q^*, p^*), p(q^*, p^*)\big) \tag{21}$$

GELFAND-P000882

Quantization

Without constraints. Let the Hamiltonian $H(q, p)$ and the action be given

$$S(q, p) = \int dt \, (p\dot{q} - H) \tag{22}$$



Original source details: GELFAND-P000882

Uncertain reading. There is a cancelled entry before $S(q, p)$. It may read $H(q, p)$.

Eqs.:

$$\dot{q} = \{q, H\} \qquad \text{and likewise for } p \tag{23}$$

Quantization: quantities are repl. by operators, and the eqs. hold

$$\dot{q} = \frac{1}{i}[q, H], \quad \dots \tag{24}$$



Original source details: GELFAND-P000882

Uncertain reading. The denominator sign in (24) is heavily retraced. It is read as i.

and the commut. rel.

$$\frac{1}{i}[q^i, p_j] = \{q^i, p_j\} = \delta^i_j \tag{25}$$



Original source details: GELFAND-P000882

Uncertain reading. A sign is cancelled below q^i in the first bracket in (25). It may be a subscript i.

Consider:

$$Z(J, I) = \langle 0 | T e^{-i \int H_{\text{ext}}\, dt} | 0 \rangle \tag{26}$$

$\langle 0|$ — ground-state vector of the syst. at the initial instant

(arb. initial vector

T — stand. T-product (arrang. the opers. in increasing time order).

Here:

$$H_{\text{ext}} = -J_i q^i - p_i I^i \tag{27}$$



Original source details: GELFAND-P000882

Uncertain reading. There is a cancelled entry at the end of (27). The final sign resembles q. The full reading remains uncertain.

(here the operators evolve in time by means of eqs. (24) —
— this is how the equation involves H).

Source eqs.. (follow from this)



Original source details: GELFAND-P000882

Uncertain reading. Above (28), a cancelled line contains $\dot{q}$ and $\{q, H\}$. The sign between them may be $=$ or $-$.

$$\left(\dot{q} - : \{q, H + H_{\text{ext}}\} :\right)\Big|_{\substack{q = \frac{\delta_r}{i\delta J} \\ p = \frac{\delta_\ell}{i\delta I}}} Z = 0 \tag{28}$$

$: \dots :$ — all sources are placed to the left of the differentiation opers.

[Below “all sources”: I and J.]

Eq. (28) is solved using a Fourier transf.:

$$Z = N \int [dq][dp] \exp\left\{ i \int dt \, (p\dot{q} - H - H_{\text{ext}}) \right\} \tag{29}$$

N — normalizing constant.

[In the right margin:] T-prod. is underdefined (at the same instant it does not distinguish opers.)

With 2nd-class constraints.

Dirac bracket $\rightsquigarrow$ commutator.

one must specify equal-time commut. rel.

GELFAND-P000883

[source number: 40]

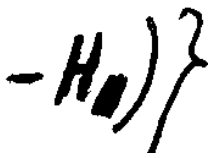
$$\frac{1}{i}[q^i, p_j] = \{q^i, p_j\}_D = \delta^i_j - \{q^i, T_\alpha\}(Q^{-1})^{\alpha\beta}\{T_\beta, p_j\} \tag{30}$$

(31) Eq. (28) is the same, with the Dirac bracket in it

It is solved again by the Fourier method:

Answer:

$$Z = \int [dq][dp] \exp\left\{ i \int dt (p\dot{q} - H_0) \right\} \prod_t \delta(T(q,p)) \big(\text{Det}_{\pm}\{T,T\}\big)^{1/2}, \tag{32}$$



Original source details: GELFAND-P000883

Uncertain reading. The subscript on H in (32) is inked over. It may be 0.

Here:

[On the right:] δ — $(2m)$-dimensional δ-fn.

$$H = H_0 + H_{\text{ext}} \tag{33}$$

On 1st-class constraints.

Let there be constraints T'_α, $\alpha = 1, \ldots, 2m$, of the 1st class

T'' — 2nd-class constraints.

The action

$$S = \int dt(p\dot{q} - H)\Big|_{T=0} \tag{34}$$

is invariant under transformations

$$\delta^F q = [q, T'_\alpha]F^\alpha \tag{35}$$

$F^\alpha = F^\alpha(t)$ — parameter!

This is gauge inv. in the Ham. formalism.

The constraints do not uniquely fix the dynamics.

Choose a repr. from each class — gauge fixing!

Add new constraints:

$$T'_\alpha = 0, \qquad \Phi_\alpha = 0 \tag{36}$$

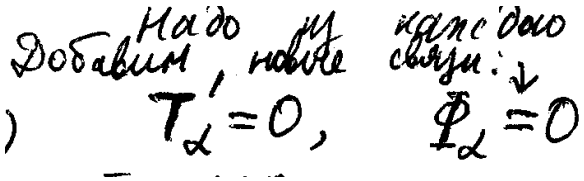


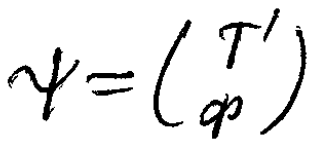
Diagram and typed description.
new constraints:
↓
$\Phi_\alpha = 0$

Original source details: GELFAND-P000883

such that

$$\text{Det}\{\Phi, T'\} \neq 0 \tag{37}$$

$$\psi = \begin{pmatrix} T' \\ \varphi \end{pmatrix} \quad \text{— 2nd-class constraint!}$$

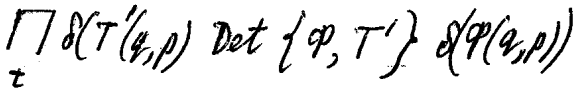


Uncertain reading. A psi sign precedes the column. Both ψ and Ψ are possible.

Original source details: GELFAND-P000883

The answer is already written down

$$Z' = \int [dq][dp] \exp\left\{ i \int dt (p\dot{q} - H) \right\} \prod_t \delta(T'(q,p)\, \mathrm{Det}\{\varphi, T'\} \delta(\varphi(q,p))$$



Uncertain reading. One closing parenthesis is visible after $T'(q,p)$ and before Det.

Original source details: GELFAND-P000883

Independence of φ is traced using BRS symmetry.

GELFAND-P000884

19.11.79. Gelfand Sem..

Where do dilogarithms occur?

1) K-theory (Bloch, Beilinson.)

2) combinat. theory of char. classes (G.-G.-L.)

3) volume of a non-Euclidean tetrahedron (Gauss, --)

4) $\zeta(3)$ (Apéry)

5) Webs (Bol et al.)

[To the right of item 1:] 6) Scattering theory

Dilogarithm:



Uncertain reading. The end of the heading is heavily retraced. It may be singular or plural.

Original source details: GELFAND-P000884

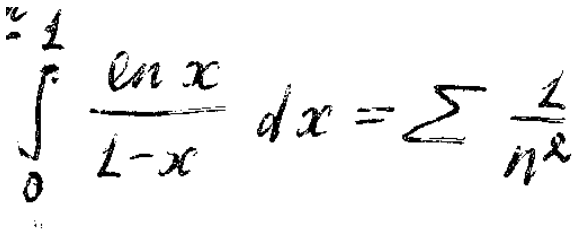
$$\int_0^{\xi} \frac{\log x}{1-x}\, dx \tag{1}$$

$$\sum_{j=1}^{5} \left(\frac{\log u_j}{1-u_j} + \frac{\log(1-u_j)}{u_j} \right) du_j = 0 \tag{2}$$

GELFAND-P000885

Beilinson

$$\int_0^1 \frac{\ln x}{1-x}\, dx = \sum \frac{1}{n^2} \tag{3}$$



Uncertain reading. No minus sign is visible before the sum in (3).

Original source details: GELFAND-P000885

Generalization to curves

(Bloch)

On the right $\zeta(2) \rightsquigarrow \zeta$ is repl. by the curve's ζ-fn.

X — Riem. surface over $\mathbb{C}$ (not necess. compact)

f, g — invertible holom. fns. on X. Consider the form:

$$\omega_{f,g} = \ln|f|\, d(\arg g) - \ln|g|\, d(\arg f) \tag{4}$$

Prop.:

$$1)\ d\omega_{f,g} = 0 \Rightarrow \text{integration along any path is well-defined} \tag{5}$$

$$2)\ \omega_{f_1 f_2, g} = \omega_{f_1, g} + \omega_{f_2, g} \tag{6}$$

$$3)\ \omega_{f,g} + \omega_{g,f} = 0 \tag{7}$$

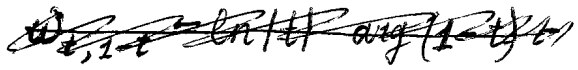
$$4)\ \omega_{f,1-f} \quad \text{is exact if } f \text{ and } 1-f \text{ are never equal to } 0 \tag{8}$$

(Steinberg ident.)

$$X = \mathbb{C}P^1 \setminus \{0, 1, \infty\}$$

$f = t$ — stand. parameter.

Then



Uncertain reading. The cancelled line reads $\omega_{t,1-t} = \ln|t| \arg(1-t)$, followed by illegible signs.

Original source details: GELFAND-P000885

$$\int_0^\xi \omega_{t,1-t} = \ln|\xi| \arg(1-\xi) - \operatorname{Im} \int_0^\xi \ln(1-t)\, d(\ln t) \tag{9}$$

The right-hand side is a single-valued fn.!! — one can show this

D. Wigner devised it and connected it to cont. cohomology of $SL(2)$.

(son)

5) Consider $\circledcirc$, going around one of the punctures x. Then

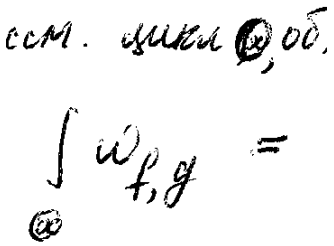
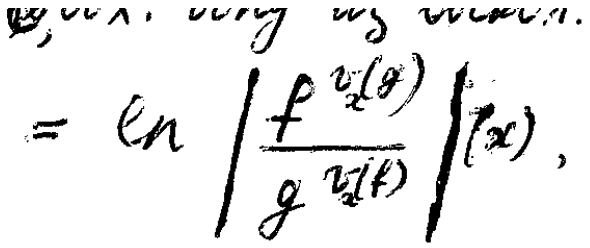


Diagram and typed description.

The interior mark resembles x. The orientation is unclear.

Original source details: GELFAND-P000885

$$\int_{\circledcirc} \omega_{f,g} = \ln \left| \frac{f^{v_x(g)}}{g^{v_x(f)}} \right| (x), \tag{10}$$



Uncertain reading. No factor is visible before the logarithm in (10).

Original source details: GELFAND-P000885

$v_x(g)$ — order of the zero of g at the point x.

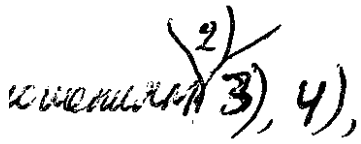
(an alg. curve is compactified uniquely. E.g., embed it into proj. space and take the closure)

This gives a map

$$K_2(X) \to H^1(X, \mathbb{R}) \tag{11}$$

f, g — fns. on X, vanishing nowhere.

Consider $\{f, g\}$ — as a free group and quotient by the relations 2), 3), 4),



Uncertain reading. "2)" is inserted above the line with an insertion mark before "3)".

Original source details: GELFAND-P000885

GELFAND-P000886

[source number: 47]

with the relations of bimultiplicativity, etc.

$\omega_{f,1-f} = 0$ in $H^1(X, \mathbb{R})$, i.e. $\{f, 1-f\} = 0$.

This is $K_2(X)$ of the field of rat. fns. on the curve.

$K_2(X)$ Define using the exact sequence:

$$\begin{aligned} K_2(X) \to K_2(\mathbb{C}(X)) &\to \bigoplus_{x \in X} \mathbb{C}^* \\ \{f, g\} &\longrightarrow \sum_{x \in X} (-1)^{v_x(f) v_x(g)} \frac{f^{v_x(g)}}{g^{v_x(f)}} \end{aligned} \tag{12}$$

From identity (10) it follows that there is a unique well-defined map (11). Consider the lin. combination



Uncertain reading. An illegible sign is cancelled before the sum in (13). It may be φ.

Original source details: GELFAND-P000886

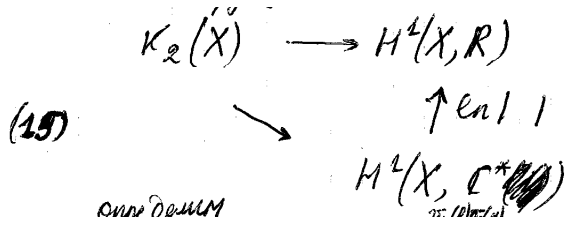
$$\sum n_i\{f_i, g_i\} \in K_2(\mathbb{C}(X)) \tag{13}$$

Discard the zeros and poles. The integrals around the loops $= 0$

$\Rightarrow$ we obt. a cohomology class in $H^1(X, \mathbb{R})$ after taking the differ.

$$\left\{\sum n_i \omega_{f_i, g_i}\right\}. \tag{14}$$

There is a construction



Original source details: GELFAND-P000886

Diagram and typed description.

$$\begin{array}{ccc} K_2(X) & \longrightarrow & H^1(X, \mathbb{R}) \\ & \searrow & \uparrow \ln | \, | \\ & & H^1(X, \mathbb{C}^* [\ldots]) \end{array} \tag{15}$$

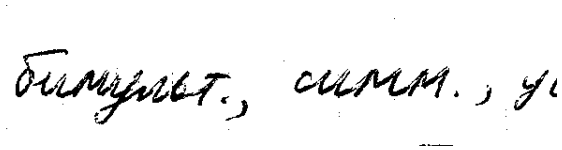
The signs after $\mathbb{C}^*$ are cancelled and illegible.

Define

$$\int_\gamma \{f, g\} = \begin{cases} (-1)^{v_\gamma(f) v_\gamma(g)} \exp \frac{1}{2\pi i} \int_\gamma \left(\ln f - \frac{v_\gamma(f)}{v_\gamma(g)} \ln g\right) d \ln g, & v_\gamma(g) \neq 0 \\ \exp \frac{1}{2\pi i} \int - \ln g \, d(\ln f) \end{cases} \tag{16}$$

(Beilinson)

This symbol is bimult., symm., satisfies the Steinberg ident., and gives the preceding one upon integration.



Uncertain reading. The manuscript reads “симм.” (“symm.”).

Original source details: GELFAND-P000886

X — aff. curve $\Rightarrow$

$$K_2(X) = H_2(SL_n(\text{alg. fns. on } X), \mathbb{Z}). \tag{17}$$

$$1 \to K_2(X) \to St^X \to SL^X \to 1 \tag{18}$$

(see Milnor and Steinberg).

— · —

GELFAND-P000887

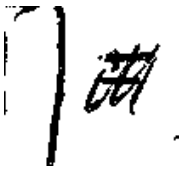
Batalin. On quantization of systems with constraints in a relativistic gauge.

$$S = \int (p\dot{q} - H_0 - \lambda T)\, dt \tag{1}$$

$$\varphi = \begin{pmatrix} q \\ \lambda \end{pmatrix} \quad \text{— relativ. field;} \tag{2}$$

In electrodyn.

$$\begin{pmatrix} \vec{A} \\ A \end{pmatrix} \quad \text{— where } A \text{ — } \lambda\text{-multiplier of the constraint.} \tag{3}$$



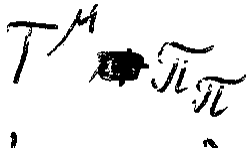
Original source details: GELFAND-P000887

Uncertain reading. An illegible sign is cancelled at the upper right of the column in (3).

In gravitation $\varphi^\alpha(q,p)$, $g^{0\alpha}$.

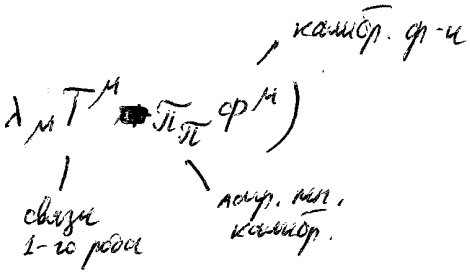
Now

$$S = \int (p\dot{q} - H_0 - \lambda_\mu T^\mu \text{ [?] } \tilde{\pi}_{\tilde{\pi}}\varphi^\mu) \tag{4}$$



Original source details: GELFAND-P000887

Uncertain reading. The sign before $\tilde{\pi}_{\tilde{\pi}}$ is heavily overwritten. It may be $-$ or $+$.



Original source details: GELFAND-P000887

Diagram and typed description.

gauge fns.

$\lambda_\mu T^\mu$ [?] $\tilde{\pi}_{\tilde{\pi}}$ φ^μ

1st-class constraints

Lagr. mult. for the gauge

$$\Phi = \Phi(p,q)$$

Relativ. inv. gauge condition

$$\Phi^\mu(p,q,\lambda,\dot{\lambda}) = -\dot{\lambda}^\mu + \chi^\mu(\lambda,p,q,\tilde{\pi}) \tag{5}$$

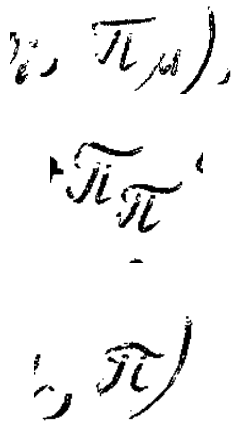


Original source details: GELFAND-P000887

Uncertain reading. The first argument of χ^μ in (5) is retraced. It may be λ.

$$\begin{cases} q^A = \begin{pmatrix} q^i \\ \lambda^\mu \end{pmatrix}, \quad p_A = (p_i, \bar{\pi}_\mu), \quad A = 1, \dots, n+m; \\ G_a = (\bar{\pi}_\mu, T_\mu), \quad a = 1, \dots, 2m; \quad \chi^a = -\begin{pmatrix} \chi^\mu \\ \lambda^\mu \end{pmatrix} \end{cases} \tag{6}$$

Uncertain reading. In (4) the subscript on $\tilde{\pi}$ resembles a second $\tilde{\pi}$. In (5) the stroke over π is wavy. In (6) the stroke over π is flatter.



Original source details: GELFAND-P000887

This is relativ. phase space. Then

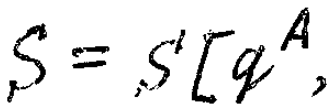
Uncertain reading. An illegible sign is cancelled before “This”.



Original source details: GELFAND-P000887

$$\begin{aligned} S = S'[q^A, p_A] = \int dt(p_A \dot{q}^A - H_{n+m}); \\ H_{n+m} = H_0 - G_a \chi^a. \end{aligned} \tag{7}$$

Uncertain reading. A short detached stroke follows the second S in (7). It is read as the prime in S'.



Original source details: GELFAND-P000887

Here the increase in the number of variables is explicit
(but in fact $(n-m)$)

$$n - m = (n+m) - 2m$$

We must suppress $2m$ extra degrees of freedom.

$$Z = \int \left(\prod_{t,A,a} dq^A \, dp_A \, d\eta^a \, d\mathcal{P}_a \right) \exp\left\{ i \int dt \left(p_A \dot{q}^A + \mathcal{P}_a \dot{\eta}^a - H_{\text{complete}} \right) \right\} \tag{8}$$



Uncertain reading. The subscript of $\mathcal{P}$ in the exponent in (8) is retraced. It is read as a.

Original source details: GELFAND-P000887

where

$$H_{\text{complete}} = H_1 - \{\Psi, \Omega\}. \tag{9}$$

GELFAND-P000888

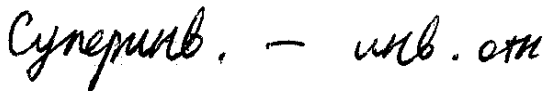
[source number: 48]

Here H_1 and Ω are found from the eqs.:

$$\{\Omega, \Omega\} = 0, \qquad \{H_1, \Omega\} = 0 \tag{10}$$

$$\Psi = \mathcal{P}_a \chi^a \tag{11}$$

Ψ, Ω — fermionic var.
If the constraints are 1st class, the eqs. are solved as series in powers of η^a and $\mathcal{P}_a$. In supergravit. these tend to terminate.
Superinv. — inv. under [cancelled] BRS-transf.:



Uncertain reading. The line begins with "Супериив." ("Superinv.").

Original source details: GELFAND-P000888



Uncertain reading. There is an illegible cancelled entry after "отн." ("under").

Original source details: GELFAND-P000888

$$\delta X = \{X, \Omega\}\mu, \qquad \mu \text{ — odd var. (constant).} \tag{12}$$

Ω — fermion
choose:

$$\mu = i \int dt \, \mathcal{P}_a \cdot \delta\chi^a \tag{13}$$

This gives BRS-inv.

One Monday is skipped.

GELFAND-P000889

3.12.79. Gelfand Sem.

Physicist. On modern theory of elem. particles.

QCD — strong 1972 (final form) Gell-Mann

$$\underbrace{\text{QED} + \text{weak}}_{\text{quant. electrodyn.}} \Longrightarrow \text{QFD}$$

1967 (Weinberg, Salam) Glashow

GR Exper. facts — the Lagrangians are correct

Faddeev, Slavnov — wrote down the Lagrangians



Original source details: GELFAND-P000889

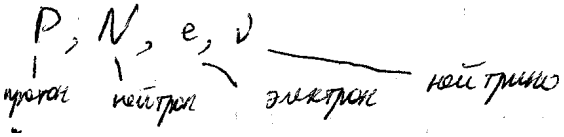
Uncertain reading. After the names the manuscript reads “выписали лагранжианы” (“wrote down the Lagrangians”).

We need:

1) group reprs.

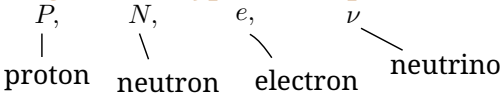
2) Feynman diag. (see Bjorken, Drell, vol. 1)

The path from the Lagrangians to exp. data is short !!



Original source details: GELFAND-P000889

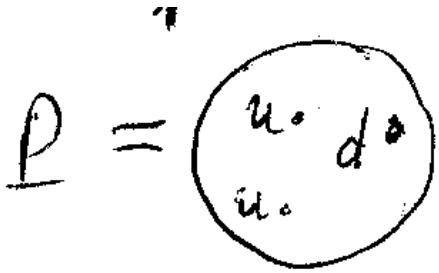
Diagram and typed description.





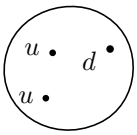
Original source details: GELFAND-P000889

Uncertain reading. A sign, possibly P, is cancelled below “proton”.



Original source details: GELFAND-P000889

Diagram and typed description.

$\underline{P} =$ 

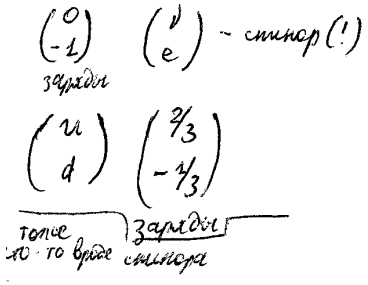


Diagram and typed description.

$\binom{0}{-1}$ $\binom{\nu}{e}$ — spinor (!)

charges

$\binom{u}{d}$ $\binom{\frac{2}{3}}{-\frac{1}{3}}$

also charges

something like a spinor

Original source details: GELFAND-P000889

GELFAND-P000890

Fractions cannot be avoided (make the charges the unit matrix)

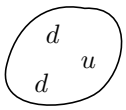
$$\binom{\nu}{e} \quad \binom{u}{d} \text{ — basic elem. particles.}$$

[On the right:] (1st generation) of particles

They suffice to construct the surrounding world



Diagram and typed description.

$N =$ 

Original source details: GELFAND-P000890

There are other generations

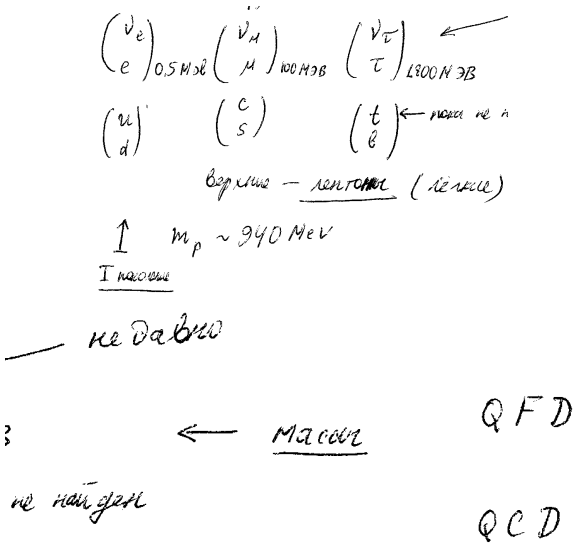


Diagram and typed description.

recently

$\binom{\nu_e}{e}$ $\binom{\nu_\mu}{\mu}$ $\binom{\nu_\tau}{\tau}$ QFD

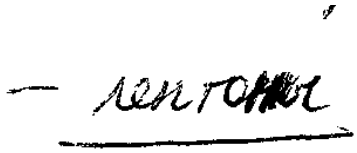
0,5 MeV 100 MeV 1800 MeV ← masses

$\binom{u}{d}'$ $\binom{c}{s}$ $\binom{t}{b}$ ← not yet found QCD

1st generation

Original source details: GELFAND-P000890

The upper ones — leptons (light)



Original source details: GELFAND-P000890

Uncertain reading. The end of “лептоны” (“leptons”) is overwritten.

$$m_p \sim 940\,\text{MeV}$$

The next generations, once produced, disappear quickly.

Leptons take part only in flavour dyn. — el.-magn. and weak interac.

(flavour dinamics)

Each quark is tripled:

u^1, u^2, u^3 and likewise for all quarks

d^1, d^2, d^3 (c, s, t, b)

Corresponding symmetry:

each quark is a representation of $SU(3)$

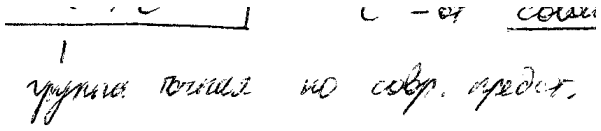


Original source details: GELFAND-P000890

Uncertain reading. The end of “представление” (“representation”) is overwritten.

$\boxed{u^1, u^2, u^3 \qquad SU(3)_c}$ C — from colour (Russian «цвет»)

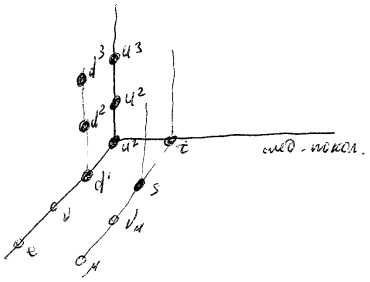
[Line from $SU(3)_c$] the group [possibly “also”] by the adj. repr.



Original source details: GELFAND-P000890

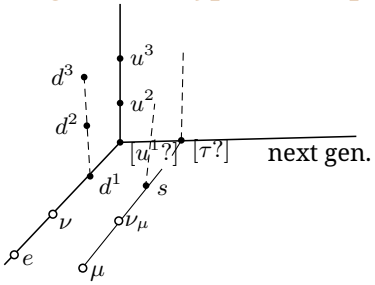
Uncertain reading. The word after “группа” (“group”) may read “тоже” (“also”).

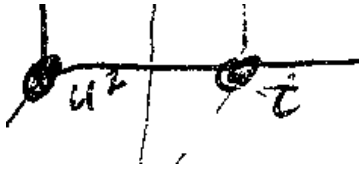
Draw the generations in a 3-dimensional diagram



Original source details: GELFAND-P000890

Diagram and typed description.





Original source details: GELFAND-P000890

Uncertain reading. The superscript on u at the vertex and the label of the right-hand point are uncertain. The right-hand label resembles τ with a detached stroke above.

[To the right of the diagram:]

$SU(3)_c$ changes the vert. comp.
(handles all verticals at once)
leptons are not tripled
(only quarks)
GELFAND-P000891

[source number: 59]

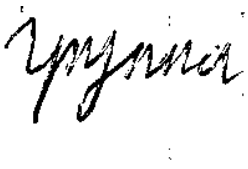
Action of the group.

$$|d^{1\prime}\rangle = U_{11}|d^1\rangle + U_{12}|d^2\rangle + U_{13}|d^3\rangle$$

U is a unitary matrix.
Or.

$$|d^{1\prime}\rangle = U_1^1|d^1\rangle + U_2^1|d^2\rangle + U_3^1|d^3\rangle$$

The leptons are permuted too!! — approximately (the masses differ)
Permutation group $SU(2)_w$ (Weinberg group)



Uncertain reading. The Russian word «Группа» is singular.

Original source details: GELFAND-P000891

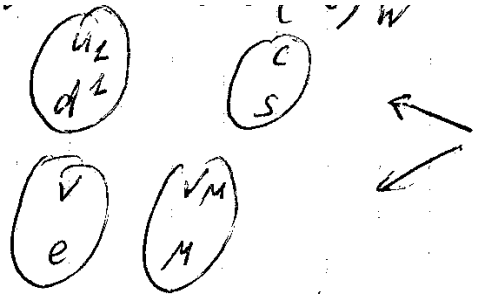
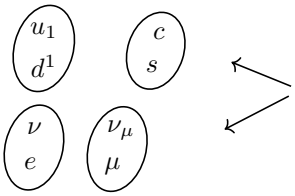
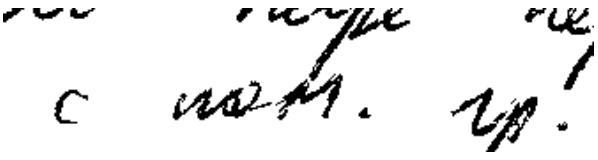


Diagram and typed description. Ovals enclosing the weak doublets, with two arrows



Original source details: GELFAND-P000891

the particles are permuted in sync within each pair by means of the group $SU(2)$



Uncertain reading. The abbreviation reads «с пом.» (by means of).

Original source details: GELFAND-P000891

We still need $U(1)$

$$[SU(2) \times U(1)]_w$$

[Crossed out:



Uncertain reading. illegible entry

Original source details: GELFAND-P000891

.] ($U(1)$ twists the phases)

Presumably e ought to have had mass 0. A transition took place as the Universe cooled.

Fermions (all those mentioned above) are described by the Dirac equation

$$(i\hat{\partial} - m)\Psi = 0 \tag{1}$$

$$\Psi = \begin{pmatrix} \psi^1 \\ \psi^2 \\ \psi^3 \\ \psi^4 \end{pmatrix}, \qquad i = 0, 1, 2, 3$$

$$\hat{\partial} = \partial^i \gamma_i = \partial_i \gamma^i, \tag{2}$$

[After the second equals sign, crossed out:



Uncertain reading. $\gamma_i \partial^i$?

Original source details: GELFAND-P000891

.]

where the γ^i are the Dirac matrices,

$$\gamma^i \gamma^k + \gamma^k \gamma^i = 2g^{ik} \tag{3}$$

$$g^{ik} = \text{diag}(1, -1, -1, -1). \tag{4}$$

Eq. (1) is Lorentz invariant

$$\partial_t = \partial_0, \quad \partial_x = \partial_1, \quad \partial_y = \partial_2, \quad \partial_z = \partial_3.$$

Degeneracy of the Dirac particle:

$$\gamma^5 = \alpha \gamma^0 \gamma^1 \gamma^2 \gamma^3 \tag{5}$$

Consider the massless eq. (1): $m = 0$

Choose α so that in the representation

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \qquad \gamma^\alpha = \begin{pmatrix} 0 & \sigma^\alpha \\ -\sigma^\alpha & 0 \end{pmatrix} \tag{6}$$

the σ^α are the Pauli matrices ($\frac{1}{2}\sigma^\alpha$ are the generators of $SU(2)$)

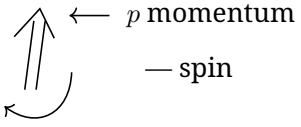
we get

$$\gamma^5 = -\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} \tag{7}$$



Original source details: GELFAND-P000891

Diagram and typed description. Momentum, spin, and the screw sense of the neutrino



$\longleftarrow$ p momentum

— spin

neutrino left-handed screw

antineutrino — — right-handed screw

GELFAND-P000892

$$\Psi_\nu = \underbrace{\frac{1+\gamma^5}{2}\Psi_\nu}_{\text{left-handed screw}} + \underbrace{\frac{1-\gamma^5}{2}\Psi_\nu}_{\text{right-handed screw}}$$

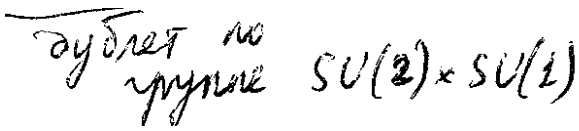
$\frac{1 \pm \gamma^5}{2}$ are projectors onto invariant subspaces

for the neutrino only $\frac{1+\gamma^5}{2}\Psi_\nu$ occurs

for the electron, both

$$\boxed{\begin{matrix}\frac{1+\gamma^5}{2}\Psi_e \\ \text{left electron} \\ \frac{1+\gamma^5}{2}\Psi_\nu\end{matrix}} \qquad \boxed{\begin{matrix}\frac{1-\gamma^5}{2}\Psi_e \\ \text{right electron}\end{matrix}}$$

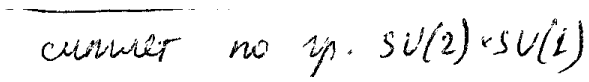
[Under the left box:] doublet under the group $SU(2) \times SU(1)$



Uncertain reading. The final factor is written $SU(1)$.

Original source details: GELFAND-P000892

[Under the right box:] singlet under the group $SU(2) \times SU(1)$



Uncertain reading. The sign between the groups looks like an incomplete $\times$ or like $<$.

Original source details: GELFAND-P000892

[Formula crossed out:



Uncertain reading. $\bar{e}\gamma^0(1+\gamma^5)\nu_e$? The superscript on the first γ and the final strokes are indistinct.

Original source details: GELFAND-P000892

]

QFD: $\gamma,\ Z^0,\ W^+,\ W^-$

Z^0, W^+, W^- are massive

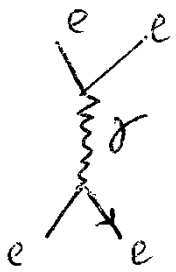
γ is light

QCD: $SU_c(3)$ 8 gluons

adjoint repr. in the Lie algebra

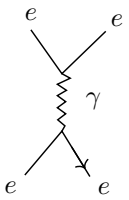
Lagrangian.

First for vector fields V



Original source details: GELFAND-P000892

Diagram and typed description. Feynman sketch with a photon



Feynman graph

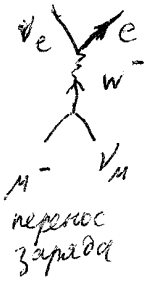
γ is the quantum [crossed out:



Original source details: GELFAND-P000892

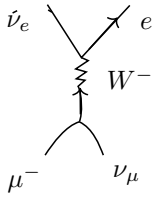
Uncertain reading. «несущ.» (carrying)?

] that transfers momentum



Original source details: GELFAND-P000892

Diagram and typed description. Sketch with W^-



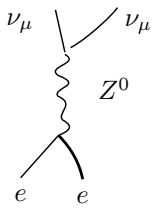
charge transfer

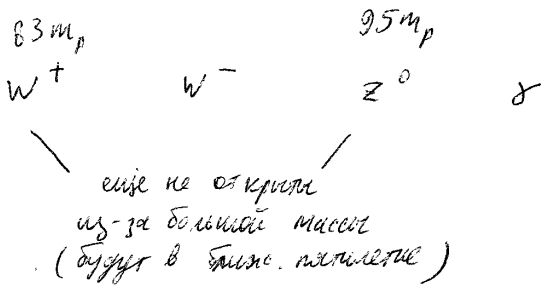
[The upper-left label resembles ν_e with an extra stroke. The mark on the adjacent line is unclear.]



Original source details: GELFAND-P000892

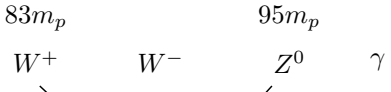
Diagram and typed description. Sketch with Z^0





Original source details: GELFAND-P000892

Diagram and typed description. Mass labels and handwritten prediction



still undiscovered
because of their large mass
(will be found in the next five years)

Charge is an eigenvalue.

pairs $\begin{pmatrix} e \\ \nu \end{pmatrix}$ are eigenvectors

GELFAND-P000893

[source number: 60]

A_a^i, a is the field number in the adjoint repr. (8)

$$F_a^{ik} = \partial^i A_a^k - \partial^k A_a^i - g f_{abc} A_b^i A_c^k \tag{9}$$

$$\mathrm{Sp}(I_a I_b) = \frac{1}{2}\delta_{ab} \tag{10}$$

I_a is an oper. in the adjoint repr. of G (basis)

Uncertain reading. The Russian abbreviation reads «опер.».

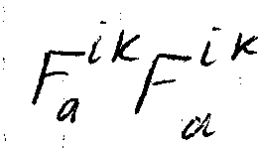


Original source details: GELFAND-P000893

Lagrangian:

$$-\frac{1}{4}F_a^{ik}F_a^{ik} + \bar{\Psi}(i\hat{D})\Psi \tag{11}$$

Uncertain reading. Both F factors are written with ik above and a below.



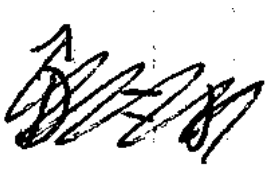
Original source details: GELFAND-P000893

Here

$$D_i = \partial_i + igI^a A_i^a \tag{12}$$

$$\hat{D} = \gamma^i D_i \tag{12}_2$$

[Crossed out at right:



Uncertain reading. $\hat{D} = \gamma$?

Original source details: GELFAND-P000893

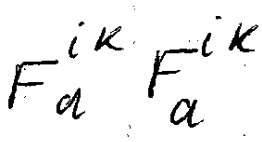
.]

Field (11) is massless

We also need scalar fields Φ (Higgs fields), transforming as a multiplet

We need:

$$-\frac{1}{4}F_a^{ik}F_a^{ik} + \bar{\Psi}(i\hat{D})\Psi + \frac{1}{2}(D\varphi)^2 - P(\varphi) \tag{13}$$



Uncertain reading. Both F factors are written with ik above and a below.

Original source details: GELFAND-P000893

Required

$$P(\varphi) = P(g\varphi) \tag{14}$$

Example: $G = U(1)$

$$P(\varphi) = \lambda^2(\varphi_1^2 + \varphi_2^2 - \eta^2)^2 \tag{15}$$



Uncertain reading. A short sloping stroke of uncertain function precedes $P(\varphi)$ in (15).

Original source details: GELFAND-P000893

(the second power is the max compatible with renormalizability)

Classical vacuum: solutions of the eqs. with min energy, i.e. those giving min $P(\varphi)$

GELFAND-P000894

10.12.79. Gelfand Sem.

The same physicist. There were 2 Lagrangians

(Igor Yuryevich Kobzar'- (?)).



Uncertain reading. The ending before the dash resembles «рь».

Original source details: GELFAND-P000894

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4}F_a^{ik}F_a^{ik}, \qquad SU_c(3) \tag{1}$$

$\mathcal{L}_{\text{FD}}$ similarly with $SU(2)\times$ [crossed out:



Uncertain reading. illegible letter

Original source details: GELFAND-P000894

] $U(1)$.

$$F_a^{ik} = \partial^i A_a^k - \partial^k A_a^i - f_{abc} A^b A^c \tag{2}$$

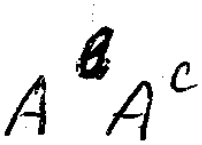
[Written separately below the last term:]

$$A^a$$



Uncertain reading. There is an additional loop at the upper stroke of $=$.

Original source details: GELFAND-P000894



Uncertain reading. The index b in $A^b A^c$ is heavily traced over.

Original source details: GELFAND-P000894

From this it is easy to compute

$$\int E^2 \, dV \sim \frac{e^2}{r} \tag{3}$$



Uncertain reading. The sign in the denominator is read as r.

Original source details: GELFAND-P000894

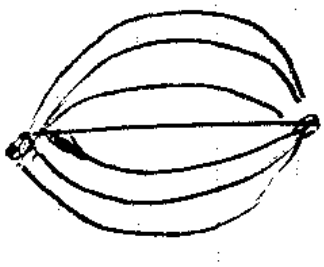
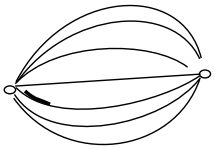


Diagram and typed description. Field lines at two small circles



Original source details: GELFAND-P000894

[A thickened mark near the left end of the inner lower arc resembles an arrowhead.]

Nonlinearity of the medium ⇒ the field lines stick together:

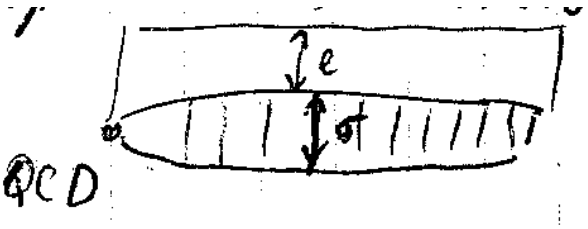
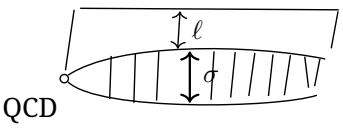


Diagram and typed description. QCD tube with a bracket and two vertical arrows



Original source details: GELFAND-P000894

[The upper label is read as ℓ.]

[To the right of the drawing:] Hypothesis: the cross-section (or diameter) of the tube is independent of the distance

min., energy density is constant $\Rightarrow$ a string results

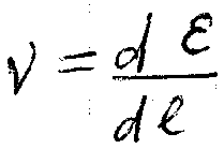


Uncertain reading. The initial abbreviation resembles «мин,».

Original source details: GELFAND-P000894

$$u^1, u^2, u^3$$

$$\nu = \frac{d\varepsilon}{d\ell} \sim E^2\sigma$$



Uncertain reading. The sign to the left of the fraction is read as ν.

Original source details: GELFAND-P000894

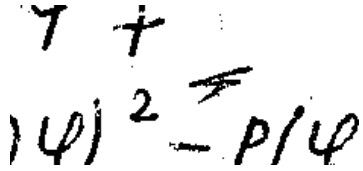
Manin's remark: when quantizing Y.–M. in the semiclassical approximation.

(and there is no exact model yet)

Weak Lagrangian:

$$\mathcal{L}_{\text{FD}} = -\frac{1}{4}F_a^{ik}F_a^{ik} + \sum_{\text{rep}} \bar{\Psi}(i\hat{D})\Psi + \\ + \frac{1}{2}(D\varphi)^2 - P(\varphi); \quad \big| (SU(2)\times U(1))_{w,s} \tag{4}$$

[Crossed out above the minus sign before $P(\varphi)$:



Uncertain reading. illegible sign

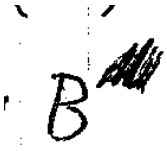
Original source details: GELFAND-P000894

.]

Here

$$D_i = \partial_i + igt^\alpha W_i^\alpha + ig'yB \tag{5}$$

[Crossed out above and to the right of B:



Uncertain reading. illegible index

Original source details: GELFAND-P000894

.]

t^1, t^2, t^3 are generators of the repr. of $SU(2)$

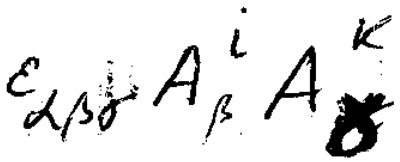
GELFAND-P000895

[source number: 66]

$$-\frac{1}{4}F_\alpha^{ik}F_\alpha^{ik}-\frac{1}{4}H^{ik}H^{ik}+\sum_{\text{rep}}\bar{\Psi}(i\hat{D})\Psi + \frac{1}{2}(D\varphi)^2-P(\varphi) \tag{6}$$

[Below the first term:] $\alpha=1,2,3.$

$$F_\alpha^{ik}=\partial^i A_\alpha^k-\partial^k A_\alpha^i-\varepsilon_{\alpha\beta\delta}A_\beta^i A_\delta^k \tag{7}$$



Uncertain reading. The third index on ε and the subscript on the final A read δ. The latter is traced over.

Original source details: GELFAND-P000895

[A separate sign above (7) on the right is crossed out:



Uncertain reading. illegible

Original source details: GELFAND-P000895

.]

$$W^{ik}=\partial^i B^k-\partial^k B^i,\qquad A_i^\alpha=W_i^\alpha \tag{8}$$

We get invariance under a local group. This was the Yang–Mills idea.

t^α, y are operators (in formulas (4))

A_α^k is a connection with values in the Lie algebra

α — $SU(2)$

k is the 1-form index

B^k is a connection with values in $U(1)$

$$\hat{D}=\gamma_i D^i \quad \text{is the standard notation} \tag{9}$$

Ψ is a 4-component spinor

$SU(2)$ is a 3-dimensional (adjoint) repr., trivial on a 1-dimensional space.

Let us examine [crossed out] the situation without vector fields: the scalar part



Uncertain reading. An illegible entry is crossed out after «Исследуем» (let us examine).

Original source details: GELFAND-P000895

$$\mathcal{L}_\Phi = [?]\,\frac{1}{2}(\partial\varphi)^2 - P(\varphi) \tag{10}$$



Uncertain reading. A small sloping sign precedes $\frac{1}{2}$. Its reading is uncertain.

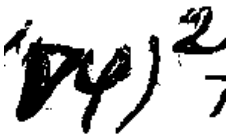
Original source details: GELFAND-P000895

φ has multiple components with respect to the group G, $\varphi = \varphi^\alpha$, $\alpha = 1, \ldots, N$
N is the dimension of the repr. corresponding to the scalar quantities.
We can assume that φ is real (if not, increase the number of components).

$$P(g\varphi) = P(\varphi) \tag{11}$$

$$T_{00} = \overset{\text{kinetic energy}}{\frac{1}{2}(\partial_t\varphi)^2} + \frac{1}{2}(\nabla\varphi)^2 + P(\varphi) \tag{12}$$

[At right:] positive def.



Uncertain reading. The sign before φ in the second term is traced over and reads ∇.

Original source details: GELFAND-P000895

min is always attained at $\varphi = \text{const}$

$$\frac{\partial P}{\partial \varphi^\alpha} = 0 \tag{13}$$

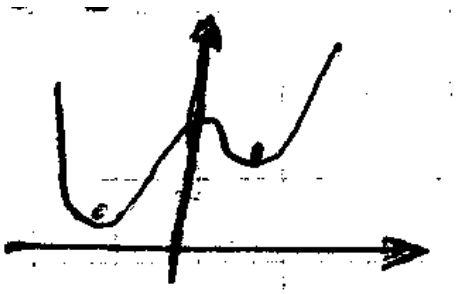
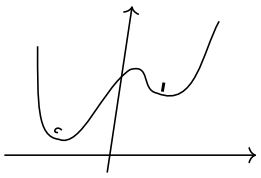


Diagram and typed description. Graph with two minima



Original source details: GELFAND-P000895

[The small marks at the minima are redrawn as strokes. The left mark resembles a letter or a point marker.]

[To the right of the drawing:] we will call any minimum (local) a vacuum.

$$\varphi^{\alpha}_{\text{vac}} = \lambda^{\alpha} \tag{14}$$

We will consider small departures from the vacuum

$$\varphi'^{\alpha} = \lambda^{\alpha} + \varphi^{\alpha} \tag{15}$$

We get the wave equation



Uncertain reading. The ending of «волновое» (wave) is written over.

Original source details: GELFAND-P000895

$$\Box\varphi^{\alpha} + \frac{\partial^2 P}{\partial\varphi^{\alpha}\partial\varphi^{\beta}}\varphi^{\beta} = 0. \tag{16}$$

GELFAND-P000896

$$\varphi^{\alpha} = c^{\alpha} e^{-ikx}, \qquad kx = kx^{i} \tag{17}$$

Uncertain reading. The second k has no written subscript.



Original source details: GELFAND-P000896

$\Rightarrow$

$$m^2 c^{\alpha} = \frac{1}{2}\frac{\partial^2 P}{\partial\varphi^{\alpha}\partial\varphi^{\beta}} c^{\beta}, \qquad m^2 = k^2 \tag{18}$$

Uncertain reading. The manuscript writes the factor $\frac{1}{2}$.



Original source details: GELFAND-P000896

Symmetry (11) $\Rightarrow$ we always get N oscillations with $m^2 = 0$



Uncertain reading. The overwritten letter in the main line resembles N. The note at right has n.

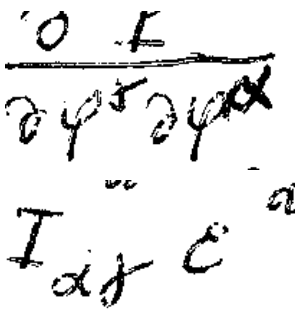
Original source details: GELFAND-P000896

[At right:] ($n = \dim G$, the number of generators, and not the dimension of the repr.)
Write the condition for invariance under infinitesimal transformations.

$$g = 1 + iI^a\varepsilon^a \tag{19}$$

$$\frac{\partial P}{\partial \varphi^\alpha} I^a_{\alpha\beta}\varphi^\beta\varepsilon^a = 0 \tag{20}$$

$$\frac{\partial^2 P}{\partial\varphi^\delta\partial\varphi^\alpha} I^a_{\alpha\beta}\varphi^\beta\varepsilon^a + \frac{\partial P}{\partial\varphi^\alpha} I^a_{\alpha\delta}\varepsilon^a = 0 \tag{21}$$

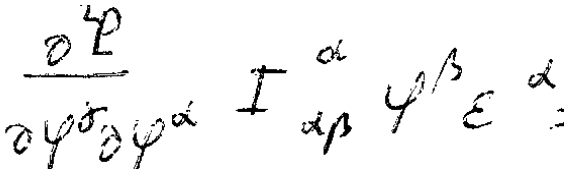


Uncertain reading. The first superscript on φ and the second subscript on I in the final term read δ.

Original source details: GELFAND-P000896

Hence.

$$\frac{\partial^2 P}{\partial\varphi^\delta\partial\varphi^\alpha} I^\alpha_{\alpha\beta}\varphi^\beta\varepsilon^\alpha = 0 \tag{22}$$



Uncertain reading. The first index in the denominator reads δ. The superscripts on I and ε are shaped like α.

Original source details: GELFAND-P000896

$$c^a_\alpha = I^a_{\alpha\beta}\varphi^\beta \tag{23}$$

Uncertain reading. The index α on c is written over.



Original source details: GELFAND-P000896

is the vector corresponding to the zero mode. (Goldstone theorem)
Some can turn out to be dependent, in general (depending on the repr.)

Uncertain reading. The initial word reads «Часть» (some).

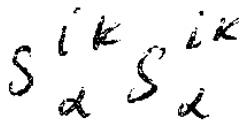


Original source details: GELFAND-P000896

$$\tau^a I^a \lambda = 0 \tag{24}$$

Consider Y.–M. on an arbitrary group G. Lagrangian.

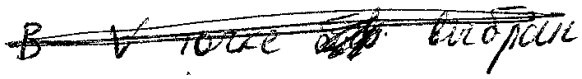
$$-\frac{1}{4} S^{ik}_\alpha S^{ik}_\alpha - \mathcal{L}(\varphi) \tag{25}$$



Uncertain reading. Both letters in the quadratic term are written as S.

Original source details: GELFAND-P000896

[Crossed out: At $\forall$ points



Uncertain reading. illegible sign

Original source details: GELFAND-P000896

we chose.]

φ^α is a scalar field (repr. of G)

$\alpha = 1, \ldots, N$

Example

$$P(\varphi) = \frac{1}{2}(\varphi_1^2 + \varphi_2^2 - \eta^2)^2, \qquad G = O(2) \quad \text{in the real repr.} \tag{26}$$

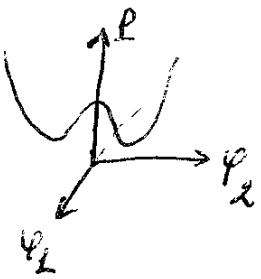
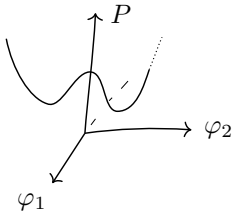


Diagram and typed description. Potential profile with three axes



Original source details: GELFAND-P000896

"bottle"

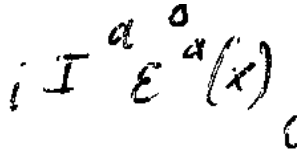
the vacuum is any state at the bottom of the bottle.

Disappearance of the Goldstone modes — a gauge is constructed.

Gauge: choice of a basis of sections, in which the connection is represented by a function

The basis of sections is constructed from the Goldstone modes. We construct rotations

$$\Phi \longrightarrow e^{iI^a \varepsilon^\alpha(x)} \Phi, \tag{27}$$



Uncertain reading. The superscript on ε resembles α.

Original source details: GELFAND-P000896

[In small writing below the formula:] and use them to arrange for Φ to be orthogonal to all the Goldstone modes.

N is the dimension of Φ

n zero modes

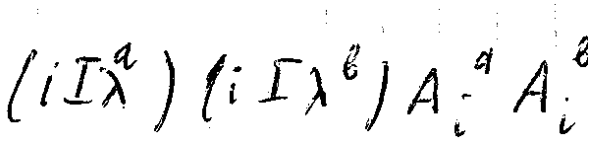
φ can be changed so that at $\forall$ points it is orthogonal to the Goldstone modes (without destroying all the properties)

GELFAND-P000897

[source number: 67]

In this procedure the particles acquired masses.

$$-\frac{1}{4}F_\alpha^{ik}F_\alpha^{ik}+\frac{1}{2}(iI\lambda^a)(iI\lambda^b)A_i^aA_i^b \tag{28}$$

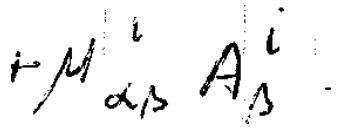


Uncertain reading. The indices a, b are written above and to the right of λ.

Original source details: GELFAND-P000897

We get

$$\frac{\partial F_\alpha^{ik}}{\partial x^i}+M_{\alpha\beta}^iA_\beta^i=0 \tag{29}$$



Uncertain reading. Both M and A have superscript i as written.

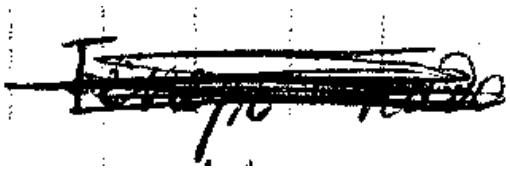
Original source details: GELFAND-P000897

(since

$$D\varphi=\partial\varphi+iI^aA^a\varphi) \tag{30}$$

This was the work of Higgs.

[Crossed out:



Uncertain reading. illegible line

Original source details: GELFAND-P000897

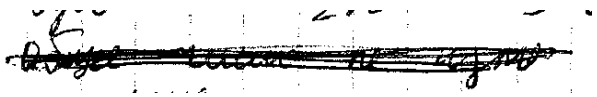
.]

The number of degrees of freedom increased to $3n$ (3 polarization dir.).

Goldstone $N \to N-n$

Here $2n \to 3n$

[Crossed out:



Uncertain reading. The start resembles «Общее» (total). The rest is illegible.

Original source details: GELFAND-P000897

]

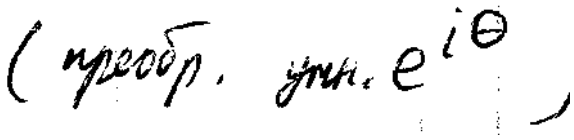
Exercise. eliminate the Goldstone modes for [crossed out] example (26)



Original source details: GELFAND-P000897

Uncertain reading. The word before «примера» (example) is crossed out and illegible.

$$A \to A + \text{const} \cdot \partial\theta \qquad \text{(transf. by mult. by } e^{i\theta}\text{)}.$$



Original source details: GELFAND-P000897

Uncertain reading. The abbreviation before $e^{i\theta}$ reads «ум.» (mult.).

Return to Lagrangian (6), fields W, B

$$\varphi = \begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} \quad \text{— spinor} \qquad \text{(4 degrees of freedom in all)} \tag{31}$$

Generators of $SU(2) \times U(1)$

$$I'_1 = gt_1, \quad I'_2 = gt_2, \quad I'_3 = gt_3, \quad I'_4 = g'y \tag{32}$$

Gauge rotations take spinor (31) to $\begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix}$

$$t_3 \begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix}$$

$$y \begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix} = p \begin{pmatrix} 0 \\ \varphi_2 \end{pmatrix}$$



Original source details: GELFAND-P000897

Uncertain reading. The coefficient and the letter in the denominators are read as p.

Hence

$$\left(2t_3 + \frac{y}{p}\right)\begin{pmatrix}0\\ \varphi_2\end{pmatrix} = 0$$

Thus $\exists$ a lin. comb. of operators that is $\equiv 0$ (depends on the vector)
(i.e. for the given state one of the Goldstone modes does not arise)

$$t_3 + \frac{y}{2p} = t_3 - y'; \qquad +y' = -\frac{y}{2p}$$

1 degree of freedom remains.
γ quantum — a vector gauging this freedom.



Uncertain reading. The heavily overwritten word reads «эту» (this).

Original source details: GELFAND-P000897

GELFAND-P000898

The mass matrix for fields $\alpha = 1, 2, 3, 4$ has the form

$$\begin{vmatrix} a^2 & 0 & 0 & 0 \\ 0 & a^2 & 0 & 0 \\ 0 & 0 & a^2 & ab \\ 0 & 0 & ab & b^2 \end{vmatrix}, \tag{33}$$

where

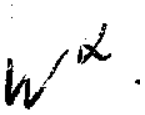
$$\begin{cases} a = g\,\text{const} \\ b = g' \cdot \text{const} \end{cases} \tag{34}$$

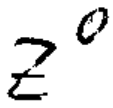
$$\frac{1}{2}\left(igt^\alpha \lambda W^\alpha + ig'y\lambda B\right)^2 \tag{35}$$

Weinberg.

$$\frac{g'}{g} = \operatorname{tg}\theta_w \tag{36}$$

$$m_z^0 = \frac{m_w}{\cos\theta_w}, \qquad Z^0 \text{ is the direction}$$





Uncertain reading. The field letters are set as W, Z. Their handwritten case is sometimes indistinct. The small indices w, z are retained.

Original source details: GELFAND-P000898

We obtain:

$$\mathcal{L}_{\text{int}} = -\frac{g}{\sqrt{2}} \left(\bar{e}_L \gamma^i \nu_L\right) W^- + \text{Hermitian conjugate} - \tag{37}$$

$$(W^- \text{ are new coord.})$$

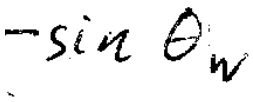
$$- g \sin\theta_w \left(\bar{\Psi} \gamma^i q \Psi\right) A_i$$

$$- \frac{q}{\cos\theta_w} \left(\bar{\Psi}_L t_3 \gamma^i \Psi_L - \sin\theta_w \, \bar{\Psi} q \gamma^i \Psi\right) Z^0$$



Uncertain reading. The numerator before the final parentheses is read as q.

Original source details: GELFAND-P000898



Uncertain reading. The final parentheses have $\sin\theta_w$ with no written exponent.

Original source details: GELFAND-P000898

Uncertain reading. One minus sign ends the first line of (37). Another begins its continuation below the annotation.



Original source details: GELFAND-P000898

q is the charge operator, $\quad q = t_3 - y$

The amplitudes of all the vertices can be read off from this, and the probability follows from it.

$$\Psi = \Psi_L + \Psi_R.$$

$$\Psi_L = \frac{1+\gamma^5}{2}\Psi, \qquad \frac{1-\gamma^5}{2}\Psi = \Psi_R$$



Uncertain reading. The spinor symbol is rendered as Ψ. Its handwritten case is indistinct.

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$\Psi \in$ tensor [inserted above: prod.] repr. of 2 spin reprs. $\times$ stand. repr. of $SU(2)$

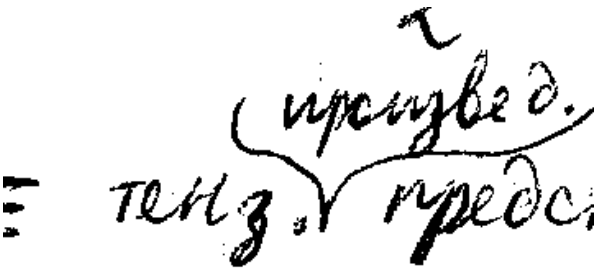
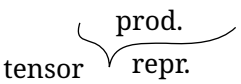


Diagram and typed description. Insertion of “prod.” between “tensor” and “repr.”



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In (37) there are currents:
1st parentheses J^-, 2nd — J^+
(charged)
3rd J^{el} — electromagn., 4th — J^0
Consider the left lepton ℓ^α_L
the right one $\bar{e}_R$
[At right:] λ_α is the vacuum expectation value of the Higgs field

$$f\,(\bar{e}_R \ell^\alpha_L \lambda_\alpha) + \text{Herm. conj.}$$

GELFAND-P000899

[source number: 68]

For a Spinor of the form $\begin{pmatrix} 0 \\ \lambda \end{pmatrix}$ we get

$$f(\bar{e}_R e_L)\lambda + \text{Herm. conj.}$$

φ — There were 4 degrees of freedom.
3 turned into Goldstones and were eaten by the vector [crossed out] mesons



Uncertain reading. The ending of «голдстоны» (Goldstones) is overwritten. A descender is crossed out below it.

Original source details: GELFAND-P000899



Uncertain reading. A word resembling «бозоны» (bosons) is crossed out before “mesons.”

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To read:

a) Faddeev, Slavnov

b) Taylor (more physical)

c) Weinberg's talk at the conf. on elementary particle physics (1974, London).

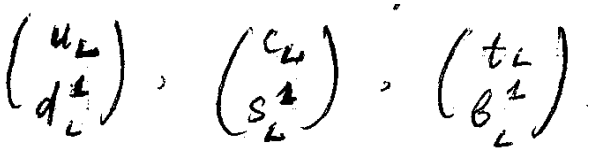
Bjorken, Drell — old.

— · —

$$\begin{pmatrix}\nu\\ e\end{pmatrix}, \qquad \begin{pmatrix}\nu_\mu\\ \mu\end{pmatrix}, \qquad \begin{pmatrix}\nu_\tau\\ \tau\end{pmatrix}$$

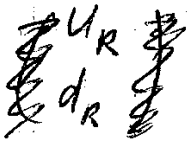
How do we write the Lagr. for quarks?

$$\begin{pmatrix}u_L\\ d_L^1\end{pmatrix}, \qquad \begin{pmatrix}c_L\\ s_L^1\end{pmatrix}, \qquad \begin{pmatrix}t_L\\ b_L^1\end{pmatrix}; \qquad \begin{matrix}u_R\\ d_R\end{matrix}$$



Original source details: GELFAND-P000899

Uncertain reading. The three doublets have superscript 1 on d, s, b. The mixing formula at right has primes.

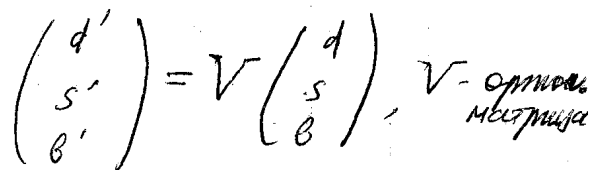


Original source details: GELFAND-P000899

Uncertain reading. Both parentheses around u_R, d_R are crossed out.

[At right:]

$$\begin{pmatrix}d'\\ s'\\ b'\end{pmatrix} = V \begin{pmatrix}d\\ s\\ b\end{pmatrix}, \qquad V \text{ is an orthog. matrix}$$



Original source details: GELFAND-P000899

Uncertain reading. The label reads "orthog. matrix."

the left ones will be doublets

the right ones are singlets

Next time: **21** January.



Original source details: GELFAND-P000899

Uncertain reading. The digit 1 in 21 is heavily traced over.